

Building A Minimally Baffled Dipole Loudspeaker

Written by: Charlie Laub, November 2022

This two-part article begins with a review of some related theory and the approach to practical design and construction of the loudspeaker. The second part describes crossover design, voicing equalization and provides a practical example built by the author.

Introduction:

I have been experimenting with and building open baffle speakers for over 10 years. What I have learned during this time has led me towards a minimal-baffle design paradigm. This delivers both benefits and design challenges. When the baffle is eliminated altogether the term "nude " is often invoked both to describe the drivers and the loudspeaker system as a whole. This is useful for distinguishing the loudspeaker from more traditional "open baffle" designs that use a baffle of greater or lesser extent. The nude driver design approach is especially advantageous at frequencies above 500 Hz where even a modest baffle can negatively influence the driver's acoustic response, on and off axis, due to diffraction from the baffle edges. A baffle can also be an undesirable secondary source of acoustic radiation as a result of rigid motion or from panel resonances, leading to coloration of the sound. The downside of little to no baffle is low frequency losses that arise because of the acoustically short distance between the front and rear of the driver's cone. I will explain how to mitigate these losses during the design process.

Part 1 – Theory and Design of the Loudspeaker

Let us begin by comparing the responses of an open baffle and boxed driver because there are both interesting similarities and significant differences. The response of a small source in the end of a finite closed cylinder is shown in Figure 1 (data adapted from reference 1). The analogous case for a similarly sized dipole source in a round open baffle is shown in Figure 2 (data from reference 2). Consider the closed box response in Figure 1. At low frequency all responses converge to 0 dB - this is the definition of a monopole; it radiates with the same magnitude in all directions. As frequency increases, the source begins to become more directional. The magnitude in the forward hemisphere increases and the magnitude in the rear hemisphere decreases. An interference pattern develops as diffraction from the edge of the baffle (the end of the cylinder) interferes constructively and destructively depending on the frequency and angle. Looking at Figure 2 we see that the dipole response, much like the closed box system, develops an interference pattern at higher frequencies as diffraction from the edge of the baffle interferes constructively and destructively. Unlike the closed box system, the response continues to fall at low frequencies and does not level off, and comparing the off-axis responses of the two systems reveals another significant difference. For the dipole, below about 1.5 times the dipole peak frequency the off-axis responses have the same shape as the on-axis response but with a reduced magnitude. Also (not shown), for a dipole the front (at an angle θ) and rear responses (at an

angle $180 - \theta$) are identical, and a null exists at $+90$ and -90 degrees (to either side). This gives rise to a great disparity in the directivity index as a function of frequency between the two types of systems. The directivity index expresses the ratio of the on-axis sound power to the sound power averaged across all axes. For a monopole the directivity index is 0 dB at low frequencies, and trends to over 10 dB at higher frequencies. The listener can easily hear this if they move off axis - the loudspeaker may sound increasingly dead away from the listening position directly in front of the loudspeaker. In contrast, a dipole source will have a nearly constant directivity index of 4.77 dB over most of its frequency range, and when equalized for flat on-axis response the directivity index remains the same. This causes a loudspeaker having a dipole type radiation characteristic to sound relatively similar regardless of listening angle.

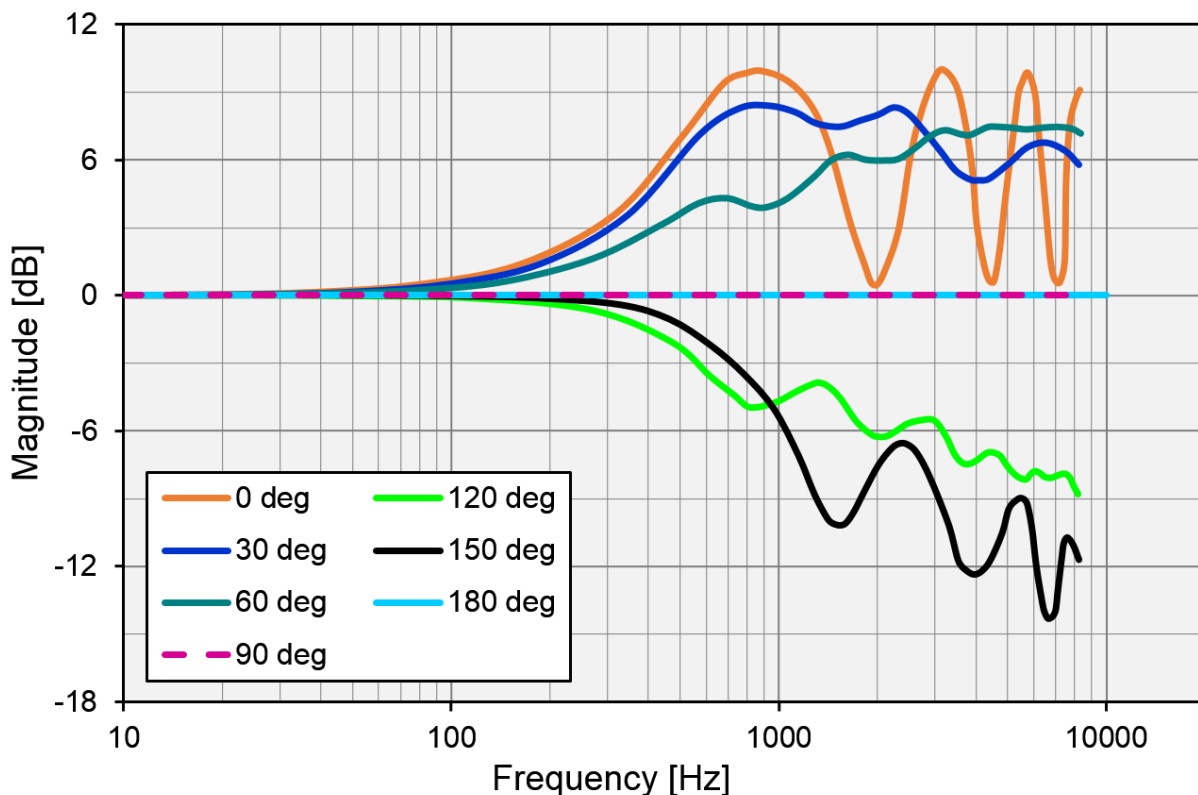


Figure 1: Response of a small source in the end of a finite closed cylinder for a selection of angles, when the baffle diameter is much larger than the driver cone.

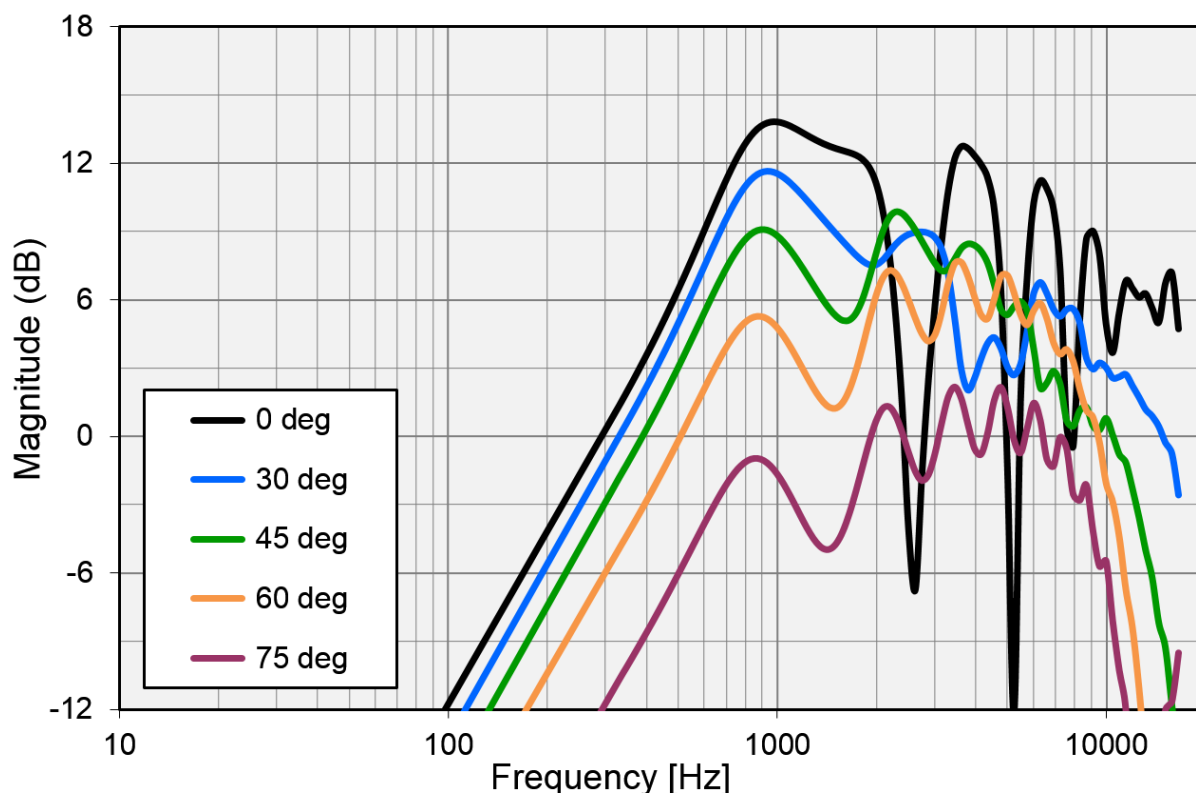


Figure 2: Response of a small source in a round open baffle for a selection of angles, when the baffle diameter is much larger than the driver cone. Data used by kind permission of Tim Mellow.

Another interesting difference between the closed box and open baffle responses is found in the vicinity of the “dipole peak”. To see this more clearly, Figure 3 overlays the on-axis responses for a dipole and closed box. The black curve corresponds to a driver in a small circular open baffle and the blue curve corresponds to the response of the same driver and baffle in the end of a cylindrical closed box. I prefer to call this response shape the “gain/loss curve”. For boxed loudspeakers this is the well-known “baffle step curve”. Figure 3 is plotted versus the dimensionless variable pair ka , where $k = 2\pi f / c$ is the wavenumber, f frequency, c the speed of sound, and a the piston radius. The dipole peak is the peak in the open baffle response that occurs at approximately $ka = 1$ of Figure 3. This peak is formed because at this value of ka the rear radiation is constructively interfering with the front radiation on axis. Between approximately one octave below and one octave above the dipole peak the on-axis response of the dipole is greater than the closed box response by approximately 3 dB. Thus, the dipole system has a higher on-axis SPL compared to an analogous closed box system across a relatively wide band of frequencies. This will play an important role in the design of the minimally baffled dipole loudspeaker.

Although there is a different mechanism responsible for the shape of the gain/loss curves for open and closed box systems, there are similarities from the point of view of the loudspeaker designer. Within the dipole gain/loss curve there is a frequency somewhere below the dipole

peak where the sound pressure level has fallen to the closed box, full-space value (0 dB in the figure). This occurs at approximately $ka = 0.4$ in Figure 3. Above this frequency, both the open and closed box response shapes follow similar trends as the wavelength becomes smaller than the baffle width. When both systems have the same ratio of baffle diameter to driver diameter the dependence on size is similar, and the use of the dimensionless variable pair ka highlights this size-frequency relationship. One way to convey the influence of size is to say that the entire gain/loss curve simply slides “leftwards” to lower frequencies as the baffle and driver increase in tandem, but the “shape” of the response remains the same. Likewise, reducing the size causes the gain/loss curve to shift “rightwards” to higher frequencies. This makes dipole and monopole systems relatively similar at mid and high frequencies.

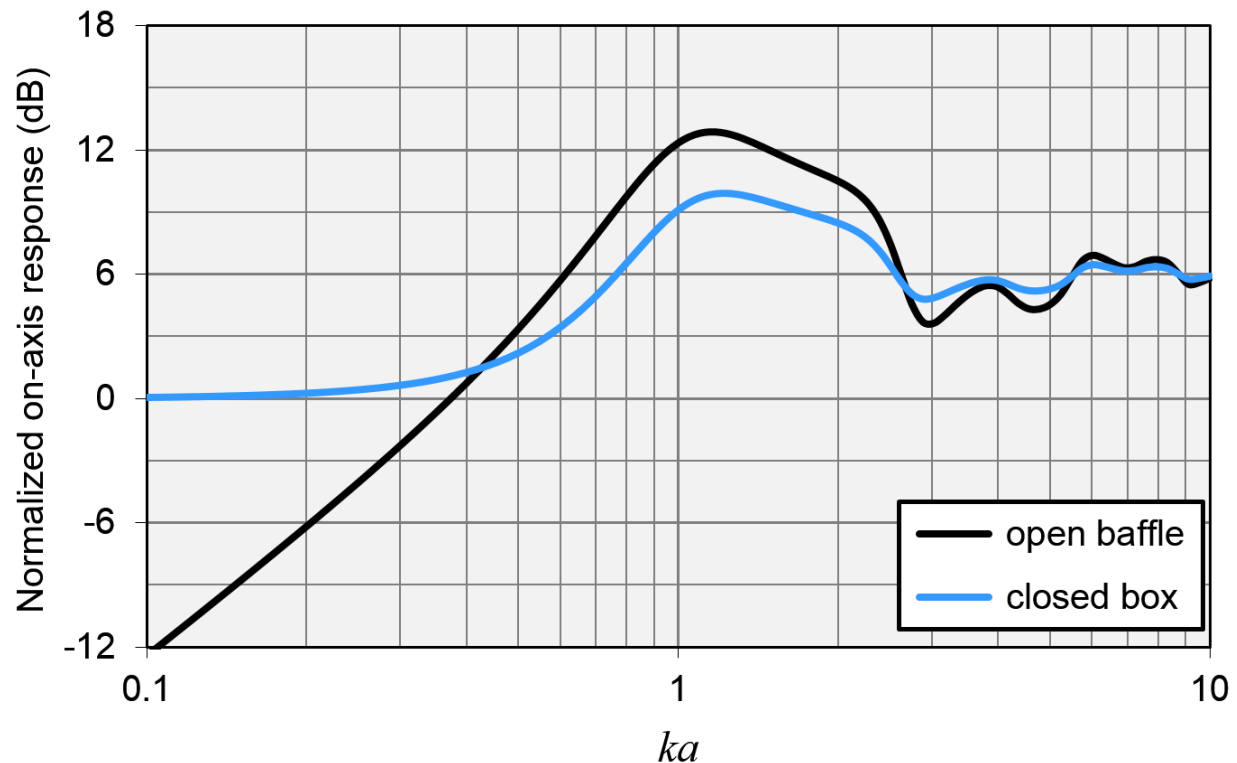


Figure 3: Example on-axis gain/loss response curves for a dipole (black) and monopole (blue) when the driver and baffle sizes are twice as wide as the driver cone. Data used with kind permission of Tim Mellow.

The steady decline in the sound pressure of a dipole source at low frequencies has consequences other than just lower acoustic output. For frequencies located well below the dipole peak, the distortion products are magnified following the rising, un-equalized driver response curve. This means that nude dipole drivers suffer from a distortion penalty compared to the same driver in a very wide baffle or with the rear enclosed. As a result, the performance requirement for drivers used as nude dipoles (midranges especially) is relatively high in order to keep distortion down to acceptable levels overall. This can be mitigated by restricting the driver to a bandwidth of no more than 2 to 2.5 octaves that is located around the dipole peak. This has consequences regarding the number and size of the drivers used in the loudspeaker.

The measured response of a real-world driver when no baffle or box is used around it has some additional features (see Figure 4). A real-world driver has finite bandwidth, meaning that it has a passband character to its output that is bounded below by the highpass character of its mechanical resonance (at F_s) and bounded above by cone breakup and rolloff at high frequencies. Just below the dipole peak the slope of the response is often faster than 6dB/octave, and this feature is not well reproduced by most dipole response models. This response “upturn” can be observed in the measured data presented in Figure 4 between 400 Hz and 1k Hz. Most drivers have additional response anomalies in the vicinity of the dipole peak, for example the peak at approximately 1.1k Hz in Figure 4. Response anomalies are often worse when the driver is smaller than the 6 inch class. On the other hand, the resonance peak may be more rounded and subtle for large diameter drivers having a very open basket.

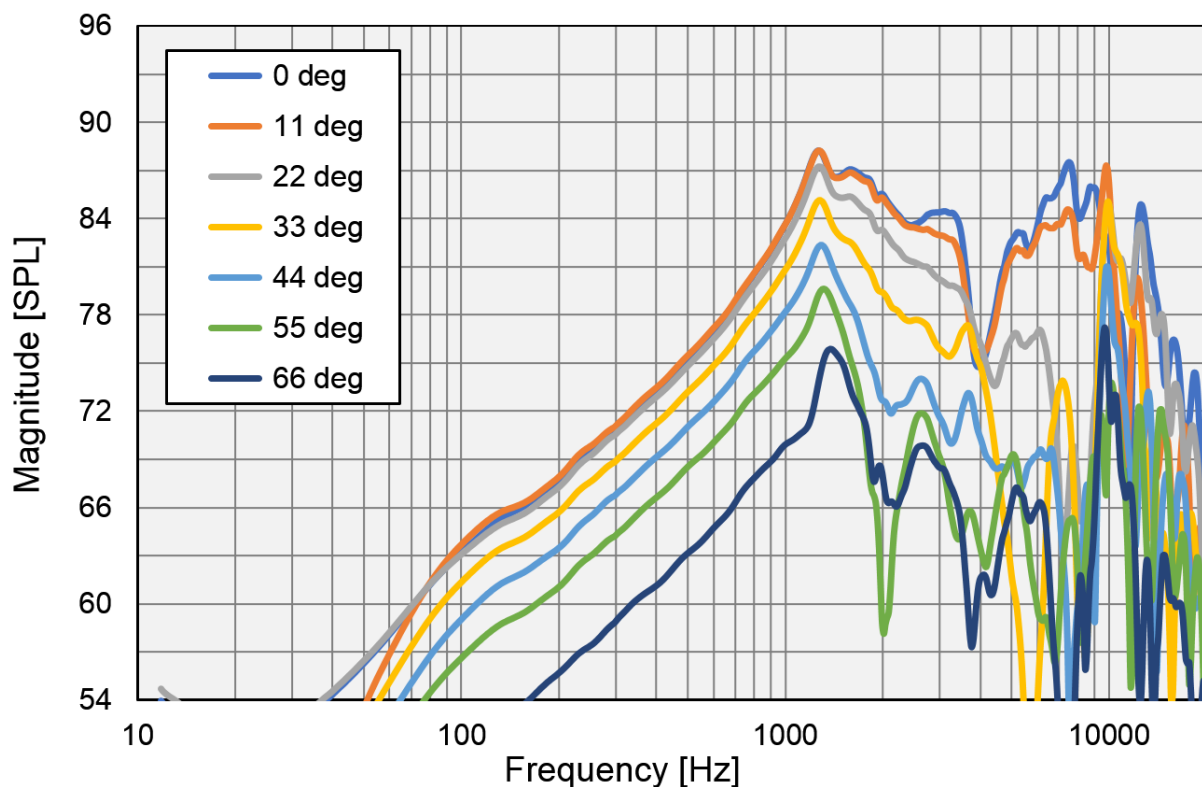


Figure 4: Quasi-anechoic response of a nude 17cm driver (from top to bottom) measured on axis and at 11, 22, 33, 44, 55 and 66 degrees off axis.

At first glance it may seem that there is an obvious solution for the high amount of low-frequency loss in a dipole loudspeaker, and that is to increase the front to back pathlength by adding a modestly-sized baffle around the driver. This can cause frequency response anomalies to appear at middle and upper frequencies that tend to reduce the usable upper limit of the driver. This was nicely illustrated by Mellow and Kärkkäinen in [2]. The authors investigated, via numerical simulation, a circular piston of fixed radius in a series of circular baffles having a ratio of one (e.g. no baffle) to eight times the piston radius as a function of the dimensionless variable pair ka . Their findings revealed that, as the baffle radius b reaches approximately twice the piston radius a , a null begins to appear in the response above the dipole peak (Figure 5, for $b=2a$,

at $ka=3$). As the baffle size increases further, the null deepens and falls in frequency. At the same time the dipole peak is moving to a lower frequency ($ka=2.1$ for $b=a$, $ka=1.1$ for $b=2a$, $ka=0.6$ for $b=4a$, and $ka=0.3$ for $b=8a$ in Figure 5) because the front to back pathlength is increasing along with baffle radius. Figure 5 shows that larger baffles reduce the loss at low frequency but also reduce the usable upper frequency.

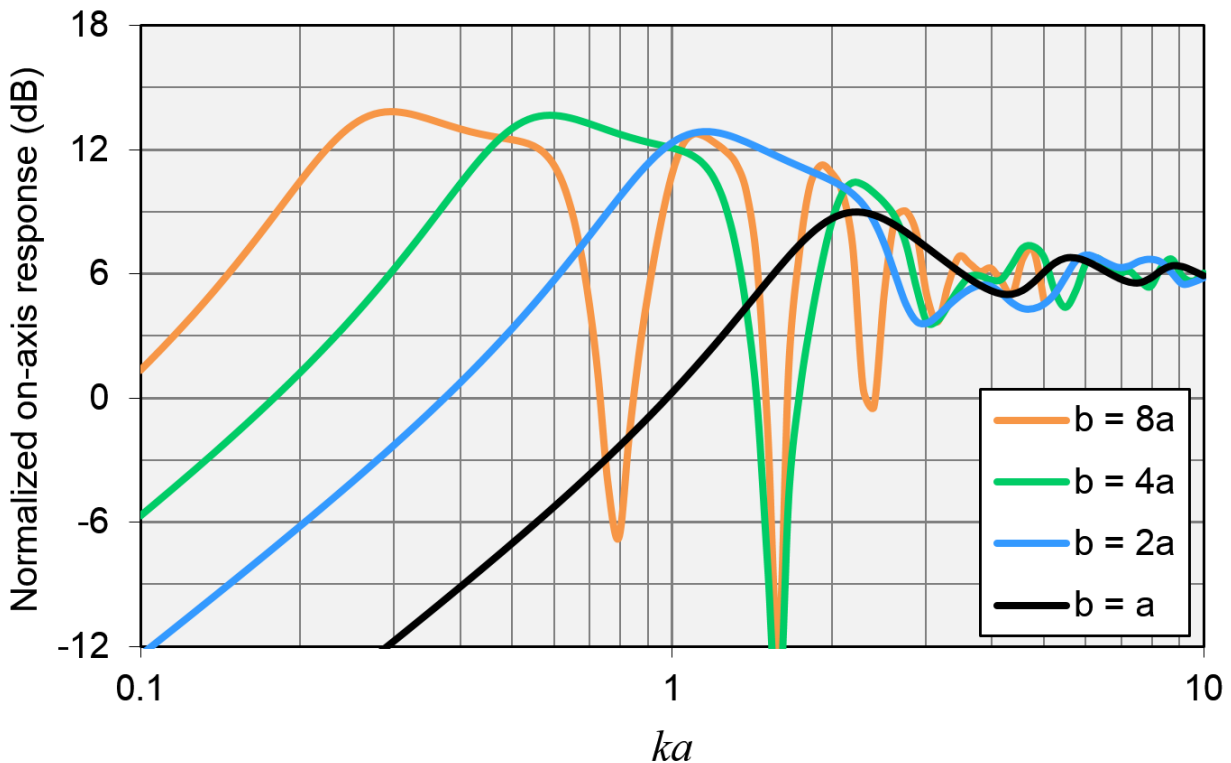


Figure 5: The on-axis response of a piston of fixed diameter a in a planar circular open baffle of diameter b as baffle radius is increased. $b=a$ (black), $b=2a$ (blue), $b=4a$ (green), and $b=8a$ (orange). Data used with kind permission of Tim Mellow.

Further information can be gained by studying the case where the baffle radius b remains fixed but the source radius a is allowed to vary. To show this, I replotted the data from reference 2, with the result shown in Figure 6. Note that the dipole peaks for each case now occur at the same frequency. This is because the baffle is the same size for each, and baffle size determines the frequency at which the peak occurs. Consider the $b=a$ curve – this represents a hypothetical “unbaffled” source, so called because the source has a diameter as large as the baffle (which is absent). Such a construction leads to the least amount of sound pressure level below $ka=5$. As the size of the source (now surrounded by baffle) shrinks, the SPL level actually begins to *increase*. When the source size reaches about half the baffle diameter there are no further significant SPL gains and, like in Figure 5, a null begins to form at about 2.5 to 3 times the dipole frequency. In addition, on- and off-axis responses cross each other (see for example Figure 2). It is important to keep in mind that, for a real-world “nude” driver not mounted in a baffle, the mounting flange itself represents a “baffle” that is approximately 1.4 to 1.5 times the diameter of the cone. When the response of the dipole source at high frequency is important, the nude driver configuration

works best because it has the smoothest on- and off-axis responses and the dipole peak occurs at as high a frequency as possible. The off-axis responses of the nude driver are smooth, but this is not because edge diffraction has been eliminated. Instead, as mentioned in [2], many paths with different lengths exist between the radiator surface and various edge locations such that the illumination of the edge is smeared out in time and space. As a result, the edge diffraction no longer causes significant ripples in the frequency response on any axis. The nude driver maximizes this effect, as demonstrated by the measured response for a real-world driver shown in Figure 4.

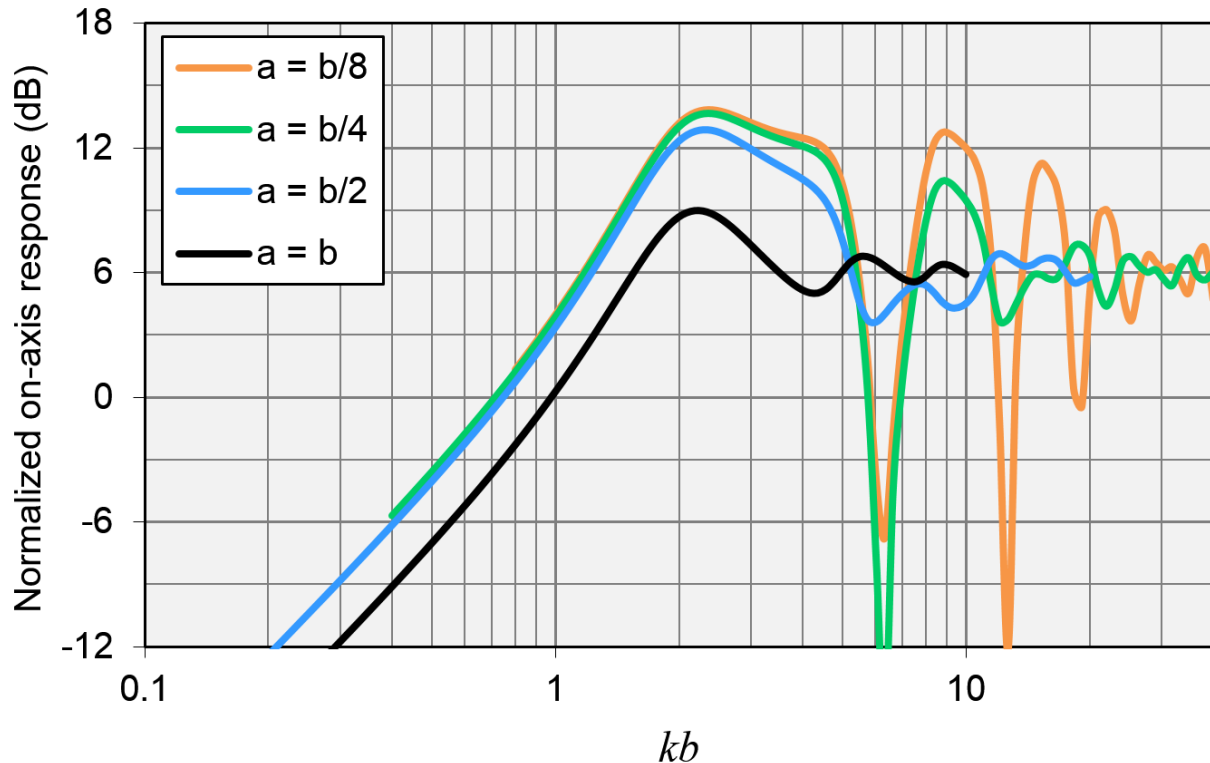


Figure 6: The response of a piston in a circular open baffle. The baffle radius, b , remains fixed while the piston radius is reduced from $a=b$ to $a = b/8$. Data used by kind permission of Tim Mellow.

Figure 5 and Figure 6 provide useful information about how to choose baffle width appropriately. When low or lower frequency performance is important, a larger driver and/or larger baffle will reduce dipole losses at the cost of moving the dipole peak lower in frequency and potentially generating dips and other response anomalies above the peak. When the driver needs to be used as high in frequency as possible it is best to use the least amount of baffle possible, which is none at all. In other words, the baffle size should depend on the role of the driver within the loudspeaker.

Even when low frequency performance is important, there is a practical maximum baffle size that is dictated by other aspects related to the acoustic performance of the dipole loudspeaker in the listening space. If the baffle is made very large and the driver offset to one side the edge

diffraction effects will be greatly diminished. Because the edge length is large this creates many dissimilar length paths from the radiator (even a point-source-like one) to the edge, and this smears out the diffraction instead of reinforcing it, just like with the nude driver. If minimizing edge diffraction was the only driver for choosing the size and shape for the baffle, a large open baffle system would seem to be a better option, however, there are other factors to consider. This sort of “infinite baffle open baffle” would not function as a dipole because little to no rear radiation would ever reach the listener except perhaps at the lowest frequencies. Part of the magic of a properly executed minimal baffle dipole loudspeaker is the soundstage, and this has been attributed to the contribution of the front wall reflection. In order to allow the front wall reflection to return to the listener unaltered, the visual size of the loudspeaker’s solid parts must be kept to a minimum. A large baffle will at least partially block the reflected rear radiation, especially if the baffle obscures the “view” of the front wall as seen from the listening position. To keep the view unobstructed, the baffle cannot be taller than about 24 inches. This allows the listener to “look over” the baffle and receive all of the acoustic energy that has been reflected from the front wall that was originally produced as the loudspeaker’s rear radiation.

Rather than employing a single large baffle to extend the operating range for one or two drivers, I prefer to use more and differently sized drivers in the loudspeaker. By increasing the number of drivers and choosing crossover points carefully, each driver can be operated in a narrower passband and closer to its dipole peak where the sensitivity is highest and the distortion penalty lowest. This also tends to maintain the dipole character of the loudspeaker’s radiation. This sort of approach is evident in other open baffle type speakers such as Linkwitz’s LX521 design. The LX521 is a 4-way loudspeaker with a minimalistic baffle. Initially he hoped that a 3-way system would lead to a wide-band dipole response but found more promise in a 4-way arrangement [3]. He arrived at the baffle shape via an iterative process in which he started out with a larger baffle and slowly made it smaller and smaller until arriving at its final shape [3,4]. In his 2017 lecture at Burning Amp, he remarked that “the best baffle is no baffle”. A video of this lecture and its PowerPoint slides are both available online [5]. This is one reason why the LX521 baffle is shaped like it is – so that it will add the minimal possible area around the drivers.

As the listener moves off axis from a properly designed dipole loudspeaker the tonal balance will remain much the same, even behind and to the side of the loudspeaker when the front wall can return all frequencies equally to the listener. When the tonal character of the reflected sound is similar to that of the direct sound it is processed by the brain in such a way as to reinforce the spatial information contained in the recording. To maximize this effect, the loudspeaker must be placed away from walls by at least 1 m and the front wall should be a plain, hard surface without diffusion or adsorption. Additionally, the frequency response to the rear must be as similar to the front as possible. Material for acoustic adsorption should be relegated to the floor, rear wall, and furnishings in order to control the space’s reverberation time. In this way the front wall can reflect all frequencies with equal intensity and high frequency information remains unattenuated. This is quite different from how most domestic listening spaces are designed. Under these conditions, the minimal-baffle dipole system will reinforce the cues contained in the stereo recording that are responsible for the illusion of depth, space, and scene. In contrast, when a loudspeaker having a frequency response that varies with listening angle is played in a room, the brain hears delayed sounds that are different in tonal character from the direct sound. The brain may believe that this is another and different sound source, and that may weaken the auditory illusion of the recording. For these sorts of loudspeakers, room treatments that rely heavily on

diffusion and adsorption are simply trying to fix the problems with the radiation characteristic that are inherent to the loudspeaker itself. The dipole loudspeaker is completely different in this regard – tonally accurate room reflections improve the sound quality rather than detract from it.

The Top-Down Design Paradigm

Now that we have covered some theory about dipole loudspeakers and baffle size, how should we design such a loudspeaker? There is a systematic approach that I call “top-down design” because the driver selection starts with the tweeter and then proceeds to the other drivers in the loudspeaker that reproduce the lower frequency bands. The requirements for each band come about as consequences of the choices made for the previous bands.

Following the reasons outlined above, the loudspeaker design should use the least amount of baffle possible and strive to have a symmetric, dipole radiation characteristic. Therefore, a tweeter capable of dipole radiation is required. Because the loudspeaker will use non-coincident drivers, the crossover point should be as low as possible in order to maintain the pattern in the vertical plane. At the same time, wavelengths are small at high audio frequencies so the size of the tweeter radiating area must also be small to prevent the directivity index from climbing excessively at high frequencies. For small dipole radiators, even the mounting flange represents a “baffle” that is too large to maintain a consistent dipole pattern. A practical solution to this problem is to remove the tweeter’s mounting flange, and when this can be done the result is often a compact dipole source with a smooth on- and off-axis radiation pattern. One example of a tweeter that can be used in this way is the Aurum Cantus AST2560, the response for which is shown in Figure 7. With the mounting flange removed, there is only a slight broadening of the pattern around 6k-8k Hz.

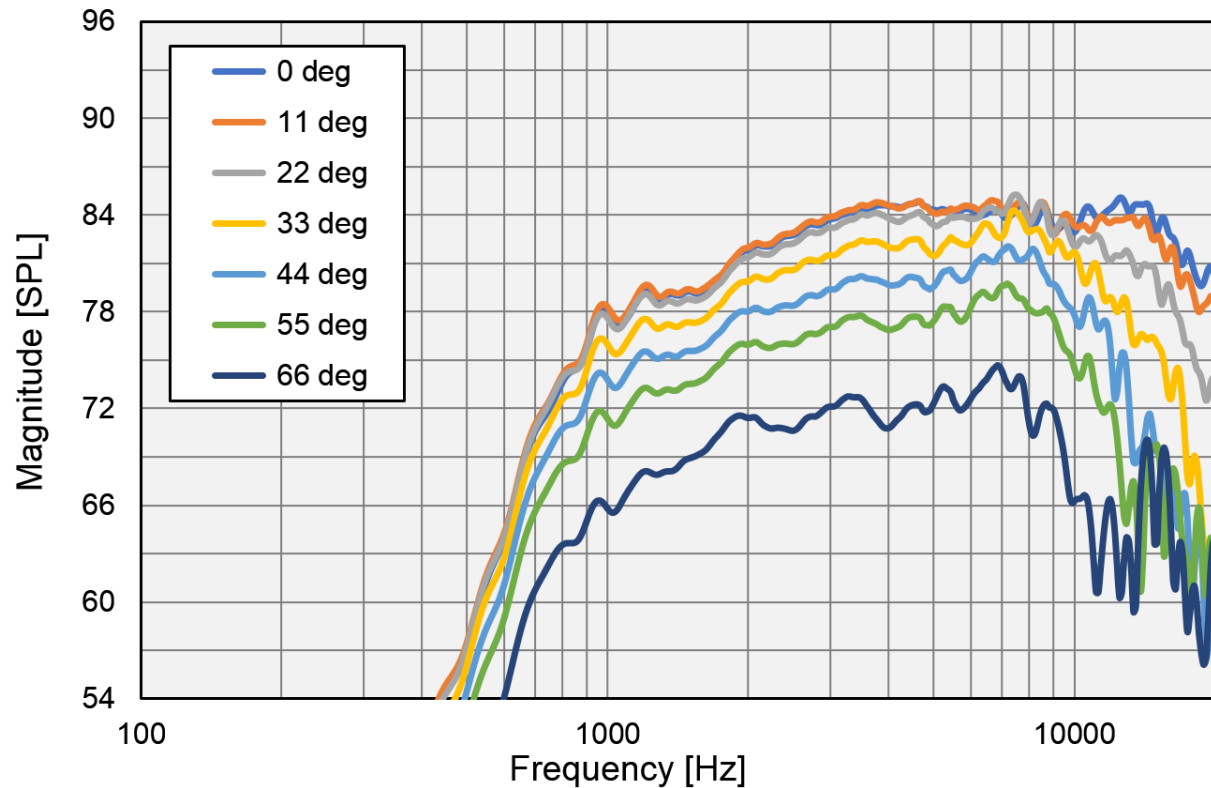


Figure 7: The response of a compact, no-baffle dipole tweeter (from top to bottom) on axis and at 11, 22, 33, 44, 55 and 66 degrees off axis.

A tweeter's lower limit of operation depends both on the frequency response and the distortion profile. Distortion will begin to rise below some frequency, and this serves as a good metric for how low in frequency any driver can be used. In general, 2k Hz is approximately the lowest useful crossover point for most dipole tweeters. Dipole tweeters that are capable of operating lower in frequency tend to be larger, and this can negatively impact the response and directivity at higher frequencies. Therefore, let us choose 2k Hz as the initial target crossover frequency between the tweeter and the next lower band.

The consequence of this choice is that we must now find a driver that can operate up to 2k Hz as a dipole, with identical output at front and rear. One option for this driver is a planar midrange, but there are currently few drivers of this type apart from the B&G Neo10, now produced as the GRS PT5010. This driver is a slim, relatively large planar driver but is limited by rising distortion and falling response below about 600 Hz when used nude, and this is not enough low frequency extension to be employed in this design. This leaves moving-coil, direct-radiator drivers. The best examples of these have lower distortion compared to planar drivers, which is important for mitigating the distortion magnification problem. The suitability of baffle-less "cone" drivers is for the most part constrained by response issues near and above the dipole peak. Drivers of 5 inch diameter and smaller tend to have strong, high-Q peaks in the rear output as low as 1k Hz, and placing a smaller driver in a baffle does not typically fix these response problems. Drivers larger than 7 or 8 inch in diameter tend to have too low a dipole peak and are

too directive by 2k Hz. The optimal driver size for this task, as borne out by measurements performed by the author on a wide variety of nude drivers, seems to be the 6 or 6.5 inch class, with very few exceptions. The response of a driver in this class that is suitable for dipole use was shown earlier in Figure 4.

Evaluating prospective drivers in the 6-7 inch class is best carried out using high resolution measurements, which must typically be performed outdoors, far away from boundaries, on- and off-axis, both in front and to the rear. Luckily this is relatively easy to do with a single, nude driver of this size compared to an entire loudspeaker. Measurements are required because little useful information about the exact response of a nude driver (a midrange in particular) can be obtained from the manufacturer's data. For example, a resonance will sometimes appear in the vicinity of the dipole peak that is not present in boxed or baffled measurements. The Thiele-Small parameters are not relevant, because the midrange will not be used anywhere near its resonance frequency. The measurement conditions used to create the manufacturer frequency response data are a poor predictor of the nude response, with the exception of the presence of breakup peaks or other high frequency anomalies. In rare cases, there are sources of public domain testing data provided by members of the DIY community that might provide a little more information especially regarding the distortion performance, but largely midrange class drivers must be tested individually specifically for their dipole characteristics, and their ability to operate up to 2k Hz and cross over smoothly to the tweeter.

Because the size of the nude midrange driver has now been chosen, we have also prescribed its dipole gain/loss curve, since that is primarily determined by the diameter of the mounting flange. In general, the dipole peak for a 6.5 inch driver will be located at approximately 1k to 1.5k Hz, and the response will roll off below this frequency due to dipole cancellation. This has important implications for the usable frequency range and subsequent input power requirements. Below the dipole peak the effective sensitivity of the driver is falling and at some point it will become too low to be practical because the input power to bring up the response to a typical moderate listening level will reach the driver's rated power. One builder remarked to me that the midrange driver (of very high quality) emanated a smell when the motor parts were heated during loud playback under the high amount of equalization that was being used. With higher power inputs comes the risk of power compression, and this may disturb the tonal balance of the loudspeaker. I try to balance the amount of boost against the desire to use the midrange as low in frequency as possible. Looking back at Figure 3 we can see that there is a frequency where the on-axis SPL has fallen to the boxed driver's free-space SPL. This will be approximately 9dB to 12dB below the dipole peak and by inspection of Figure 4 we can see that, for a nude driver in the 6 to 7 inch class, this will occur at approximately 500 Hz. Because the high-pass crossover filter will begin to attenuate power above the crossover frequency a slightly lower crossover frequency between 350 and 450 Hz can be used. This band, spanning 350 Hz to 2k Hz, is essentially the same for all drivers of this size such that the choice of driver is dictated primarily by other parameters such as voltage sensitivity, distortion performance, and the lack of resonance peaks and response anomalies.

The next task is to design the woofer section, and here the design philosophy will be slightly different. The woofer will be operated completely below the dipole peak, and the dipole loss can be significant unless the driver or baffle is large. On paper it seems that a large format pro audio driver would be especially well suited for this task, however, listening tests on loudspeaker

prototypes constructed during 2021 revealed that most PA drivers 15 inch or larger have audible energy storage or resonance problems as low as 250 Hz. The larger 18 or 21 inch class driver that would be most suitable for reducing dipole losses are worse in this regard. Instead of a single large driver the woofer section can be comprised of multiple home audio drivers of a smaller diameter (e.g. 8 to 10 inch size) mounted close together on a baffle. Unlike a single large driver, the smaller drivers have no inherent response anomalies at the upper end of the intended passband. The woofer panel approximates or exceeds the front to back pathlength of the larger single driver, which further minimizes low frequency losses. Parallel or series-parallel electrical connections raise the effective voltage sensitivity by 6dB to 9dB over that of a single driver. By locating the woofer panel on the floor, it also benefits from 6 dB of boundary gain. In addition, the frequency at which the floor acoustic reflection occurs is moved to the upper edge of the passband or just above it. In earlier nude dipole speaker prototypes, I attempted to locate the woofer higher, next to the midrange, and these designs always gave rise to a notch in the passband that was due to the floor boundary reflection. This resulted in audibly thin mid-bass that was difficult to impossible to correct in the crossover – it was a built-in design flaw. Placing the woofer at the floor largely eliminated the problem. By using a panel of modest height, the overall cross section of the loudspeaker is kept to a minimum as seen from the listening location, and by utilizing both electrical and boundary gains the woofer panel sensitivity can be made relatively high, which largely offsets the dipole losses.

Despite all the advantage of using multiple drivers mounted in a small baffle, the woofer panel should not be operated to an arbitrarily low frequency. The drivers will be operating in close to “free-air” conditions, meaning that the near-resonance response will be drooping well above F_s if Q_{ts} is not relatively high. When combined with the dipole losses, this causes the SPL loss rate to increase beyond 6dB/oct and the power required to sufficiently equalize the response increases rapidly. Just like with the midrange, low frequency distortion is magnified, and it is important to use high-quality drivers that have very low distortion below 200 Hz. Even if these problems are mitigated by using special high-excursion, high-Q drivers or larger folded baffle arrangements, dipole bass systems tend to be effective only above 30 Hz in most domestic settings. For these reasons, I choose to add another band to the loudspeaker, e.g. a subwoofer. To reduce the overall demand on the drivers in the woofer panel, it is prudent to cross over between the subwoofer and the woofer panel around 80 Hz. Most domestic listening spaces will be best served by stereo closed box subwoofers placed nearby the woofer panel and nude drivers. The radiation pattern will change through cardioid at the crossover frequency between the box subwoofer’s monopole response and the dipole response from the 3-way speakers. The total radiated power will rise in the transition band because a monopole delivers 4.77 dB higher total radiated power than a dipole system for the same on axis SPL. This makes it easier to deliver more realistic bass SPL via a monopole source than a dipole one. Closed box subwoofers have the advantage that they can pressurize the room and will bring the familiar bass impact that many feel is lacking with dipole bass systems. The familiar monopole subwoofer is something that many people may already own, know how to set up and locate in the room, and can be easily tuned to match the room gain in the listening space. I have found that an 80 Hz crossover point between the subs and woofer panel is a practical one and the transition from dipole to monopole is seamless throughout the crossover region.

The Wireframe Scaffold

With the driver selection process completed it is time to consider how to practically create the loudspeaker. If some drivers are to be “nude”, that is without any baffle or other driver touching them, how do they stay in place - suspended in the air as if by magic? In the past I have built loudspeakers with spines that supported certain drivers, or by hanging the drivers by wires from large frames that surrounded the drivers at a distance. In each of these cases, some material was acoustically “in the way” to one extent or another. A better way was needed.

The wireframe construction technique can effectively solve the challenge of creating true “no baffle” conditions and is a structure that is acoustically transparent to the driver’s radiated output. This concept took shape after I discovered a welded 1x1 inch steel wire mesh with a black polyvinyl coating. This material is DIY friendly and is very well suited for this type of construction. It is available in rolls and is intended for fencing, etc. The 14-gauge wire and welded construction provides stiffness and strength. The vinyl coating helps to dampen resonances in the wire itself, making it acoustically neutral. Rolls of this material are available in the USA from the Fencer Wire company (fencerwire.com) in 24 and 48 inch widths, in 50 yard or longer lengths. The characteristics of this material make it nearly ideal for a scaffolding from which the drivers can be suspended, however, it must be crafted into a shape that is appropriate for a loudspeaker.

The most practical shapes for a scaffold formed from this material have a cross-section in the shape of a “D” or triangle “ Δ ”. To create these shapes, panels of the wire mesh are cut from the roll using flush-cut pliers and trimmed so that no wire stubs/sprues remain. To make a scaffold having the “D” cross-section, one panel is flattened into the “I-shaped” face, while a larger one forms the curved “C-shaped” section. Attaching panels to each other is accomplished using small, inexpensive, 3-fingered metal clips marketed as “wire cage buckle clips” that may be purchased online [6,7]. These are similar to but slightly different from “upholstery stay wire clips” “or “edge wire clips” used in the furniture making industry. When crimped around two wires they form a compact and flush wrap that physically connects and joins them. Using dedicated crimping pliers and the clips, the panels are joined by crimping a clip every few inches between the wires that form the edges of each panel. The size of the panels determines the final cross-sectional dimension of the “D” and therefore the largest size of driver that can be accommodated inside it. To form a scaffold having a triangular cross section, three flat wire mesh panels are prepared and joined together along their edges in a similar fashion. In each case, because the drivers are to be suspended inside the wireframe scaffold, it must be large enough to accommodate them with some extra room. The midrange driver can be suspended simply by hanging wires off of the wireframe itself. Monofilament nylon such as fishing line, which has low stretch and is available in ratings up to 200 pounds or more depending on the line cross section, provides a convenient way to do this. One way to create an attachment on the end of the line is to use 14-16 gauge solderless crimp-on wire “forked” terminals. Monofilament line that is rated for 150-200 pounds has a diameter that is approximately the same as 16-gauge wire and the terminal can be securely crimped around the nylon line just like wire. The “fingers” of the forked terminals can be bent over to create a hanger for the driver at a convenient place on the wireframe. This works very well for the midrange driver. Most of the mass in a moving-coil “cone” driver is located in the magnet/motor, putting its center of mass far behind the mounting flange. Instead of attempting to attach to the driver directly, two long machine screws or bolts are

securely tightened to the holes in the mounting flange. The length of these bolts must be sufficient such that the center of mass intersects them back near the top plate. Two hanging lines of equal length are prepared - a ring terminal large enough to fit around the screw or bolt is crimped onto one end of the nylon wire, and a forked terminal crimped onto the other end. For each line, the ring is slipped over the back of the screw/bolt and aligned with the center of mass and then the bent forked terminals are hooked over the wireframe scaffolding. The driver will be hanging freely and perfectly balanced. The only remaining physical connection is the speaker wires. These can often be routed in a curved path that provides little resistance to the driver swinging on the wire. Alternately, a length of braided tinsel lead wire can be soldered on to the end of the speaker leads to form the last few inches of the connection. The braided construction makes this type of wire very flexible, and it will not offer any resistance to movement.

Hanging the midrange driver in this way has several benefits. The suspension system does not present an acoustically significant obstacle. Wires having diameters on the order of 1-2 mm are acoustically invisible yet can support a significant amount of weight. The driver is a mass at the end of the hanging lines, which together form a pendulum. The natural frequency of a pendulum ($T=2\pi(L/g)^{0.5}$) is independent of the suspended mass and depends only on the length of the wire. A line that is only a few inches long has a period that is on the order of one second. If the passband frequencies are far above the natural resonance frequency of the pendulum, the suspension decouples the driver from the wireframe scaffold. The driver mounting point is easy to relocate by sliding the forked terminals along the wireframe, and this allows the midrange to be positioned forward and back as needed in order to adjust the position of its acoustic center. Finally, the suspension system is quick and easy to put together using inexpensive and readily available parts.

The frame of a decoupled, freely hanging driver will move in opposition to its cone movement under the effect of action-reaction. When the driver is hanging freely, the counter movement of the frame is completely linear and causes a reduction in the acoustic output generated by the cone. One concern that has been voiced to me is that the level of this reduction will be significant, so I will provide a brief analysis to address the issue. Newton's Second Law states that acceleration depends on the force applied and the mass of the object ($F=m*a$) and Newton's Third Law states that for every action there is an equal reaction, that is $F_1 = -F_2$. The force F being applied is generated by the motor and this force pushes the cone against the other driver parts (frame, motor, etc). Since the driver parts are decoupled from the ground by the suspension system and are free to move, the motor force induces motion in them as well. Thus, we can write:

$$\text{cone_mass} * \text{cone_acceleration} = - \text{driver_mass} * \text{driver_acceleration}$$

Because the direction of the driver acceleration is opposite that of the cone acceleration, the negative sign can be dropped from the term on the right side and acceleration taken to mean the magnitude of the acceleration henceforth. Rearranging this we get

$$\text{driver_acceleration} = \text{cone_mass} * \text{cone_acceleration} / \text{driver_mass}$$

In this analysis, "driver" refers to everything except the cone, voice coil former, voice coil, and air load. Because the driver is accelerated away from the cone in the opposite direction, this

reduces the net acceleration of the cone compared to when the driver is immobilized by a connection to the ground via a cabinet or baffle. The net cone acceleration will then be:

$$\text{net_cone_acceleration} = \text{cone_acceleration} - \text{driver_acceleration}$$

We define the *loss_ratio* as the ratio of the cone's net acceleration to the acceleration without the loss from the driver acceleration:

$$\text{loss_ratio} = (\text{cone_acceleration} - \text{driver_acceleration}) / \text{cone_acceleration}$$

Substituting the expression for *driver_acceleration* and rearranging we get the result that:

$$\text{loss_ratio} = 1 - (\text{cone_mass} / \text{driver_mass})$$

Because the sound pressure in the far field is proportional to cone acceleration, the value calculated by the loss ratio also predicts the loss in far field sound pressure. Let's do the calculation for a lightweight neo magnet pro driver, the Eminence Deltalite II 2515: the driver weighs 2590 grams overall and the mass of the moving parts and the air load (Mms) is 79g resulting in a mass ratio of 36:1. Plugging in this ratio, we have $\text{loss_ratio} = 1 - 1/36 = 0.972$, which is approximately -0.25 dB. As an example of a home-audio 6" class driver, the SB Acoustics SB17MFC weighs 1550 grams overall and Mms is 12.3g. This represents a mass ratio of 126:1, which corresponds to -0.07 dB. These examples show that for both hi-fi and pro-audio drivers this source of sound pressure loss is typically much less than 1 dB and can be considered negligible. This analysis also highlights that the frame of the driver undergoes acceleration. It is useful to decouple it as much from other parts that can move, other drivers, or even the loudspeaker cabinet for boxed speakers since the induced vibrations will lead to cabinet panel resonances. For example, should the midrange and tweeter be affixed together but otherwise free to move, the tweeter's position will undergo motion induced by the midrange's reaction force, which would modulate the tweeter output and could cause doppler distortion. By decoupling drivers from one another within the wireframe scaffold this source of distortion can be eliminated.

Mounting the tweeter within the wireframe is a bit more straightforward because there is no appreciable motional energy generated by the tweeter and it can be mounted directly to the wireframe. But since the metal in the wireframe is steel, it will be magnetically attracted to the stray magnetic flux from the motor. To prevent such an interaction, the tweeter can be mounted to an open aluminum mesh that has been manufactured by perforating a thin aluminum sheet. This material is available from several sources [8,9,10]. Since aluminum is non-magnetic the driver is unaffected. The aluminum is run between two additional braces/beams that are located at the top of the wireframe, running front to back and spaced not much wider than the width of the midrange mounting flange (see Figure 8). In addition to supporting the tweeter, the beams provide practical hanging points for the midrange. It's not typically possible to hang the midrange driver from the main wireframe because the angle of the hanging lines would be too far away from vertical. The image in Figure 8 shows the beams/braces from which the midrange is hung, and the perforated aluminum mesh to which the tweeter is mounted, for a prototype loudspeaker utilizing the wireframe scaffold.

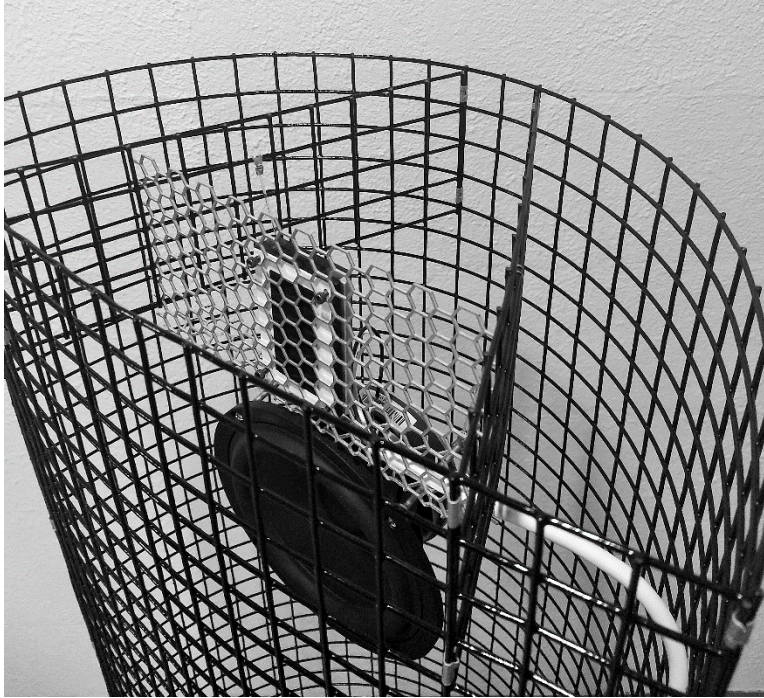


Figure 8: Upper part of wireframe showing beams and aluminum mesh that supports the tweeter.

It is important to keep objects of any size away from the main dipole lobe of the drivers, front and rear, and this includes wiring bundles. It's preferable to place wiring at 90 degrees off axis, in the partial null where reflected energy will be minimized. The most congested area is between the midrange and tweeter, where four wires typically come in and connect to the terminals. The tweeter is often fixed to a part of the wireframe directly, with the midrange suspended just below it. Routing a 4-conductor bundled wire up the side and then across to the drivers allows for a compact wire layout with minimum footprint. This can be seen in Figure 8.

In summary, a systematic design approach has been described that gives rise to a 4-way loudspeaker system consisting of a 3-way nude dipole loudspeaker plus subwoofer(s). A compact dipole tweeter operates above 2k Hz. A 6 to 7 inch driver is used for the midrange band between approximately 350 Hz and 2k Hz, and multiple home audio drivers in a minimal baffle are used from 80-350 Hz. The tweeter and midrange sit within a wireframe scaffold directly above the woofer panel, and all drivers are physically decoupled. The woofer panel sits on the floor to maximize boundary gain and is no more than 24 inches tall. By using multiple sources in the panel, the "floor bounce" effect is minimized. This 4-way design will have a nearly constant directivity index above 100 Hz and can provide excellent sound quality when the drivers are chosen correctly.

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Part 2 –Measuring the Drivers and Developing the Crossover

Crossovers for open baffle loudspeakers often require more corrections than boxed loudspeakers and this is even more so for a nude dipole system. The crossover must flatten the responses, primarily of the midrange and woofer, EQ out any peaks in the response (typically in the midrange), implement crossover filters that define the band limits for each driver, and match the output levels of the drivers. The crossover is designed as the combination of a flat on-axis frequency response plus a voicing EQ curve. An example loudspeaker design that follows this approach is described.

Clean, high-resolution acoustic measurements are essential for creating an accurate model of the loudspeaker in the crossover design software package. Unfortunately, it is not always feasible for the loudspeaker builder to make such measurements directly, especially at low frequency. Out of practicality some measurements that include at least the effect of one room boundary are made. These must be interpreted and adjusted accordingly. In all cases it is strongly suggested that measurements are made with a fixed time reference in order to capture accurate phase information.

Acoustic Center Balancing

Before making a set of measurements there is a preliminary step in which the midrange and tweeter are physically aligned on the wireframe scaffold. Once these drivers are physically moved to the appropriate position their acoustic output will be as identical as possible as seen from the front and rear. This is a goal for the dipole loudspeaker and is why drivers with identical forward and rear response within their passband are utilized. In a dipole speaker only a physical adjustment of the relative driver positions can achieve the desired effect. In contrast, only the forward radiation from the drivers of a boxed loudspeaker reaches the listener and misalignment of driver acoustic centers can be removed using a digital delay or compensated for using an all-pass or other filter. But this does not work for a dipole. When you delay the tweeter signal, for example, as seen from the front it is akin to “moving the driver backwards”. But when observed from the rear, the driver would apparently be moved forward by delay. This means that, if the physical acoustic center is too far forward, using delay will have the apparent effect of moving it towards the rear as seen from the front, but from the rear the effect is to move it even farther forward, which would only make the offset problem worse. A better way to think about the delay is that an impulse sent to the speaker would be moved farther back towards the amplifier in the speaker wire and has the same effect on both the front and rear outputs. In a dipole system the acoustic centers of two drivers must be physically manipulated - this is not “aligning” the acoustic centers per se, only making the acoustic center offset between any two drivers the same as seen from both front and rear.

In general, if a planar tweeter and a moving coil “cone” midrange are used the appropriate tweeter position will lie near the plane of the midrange mounting flange. For those who wish to check the alignment via measurement, a technique for measuring the relative acoustic offset of two drivers has been described previously by Jeff Bagby and David L. Ralf [11,12]. The reader is referred to these documents for more information. In general, a series of measurements is performed in front of the loudspeaker and repeated at the rear. The interference technique is then employed to determine the offsets using software. When both the front and rear acoustic center

offsets have been determined, the builder then physically moves one driver (typically the midrange) forward or backwards so that the offset as seen from the front and the offset as seen from rear become equal. Note that this will not necessarily result in either offset being zero. As long as the front and rear offsets are identical the crossover will have an identical effect at both the front and rear, which is the desired result. Keep in mind that, at 2k Hz, an error in the alignment of one centimeter will result in approximately 17 degrees of phase error.

Once the tweeter and midrange physical alignment has been performed, the next step is to obtain a set of measurements that are sufficiently “clean” to permit accurate crossover design. In this context clean implies free of significant peaks and dips due to reflection from room boundaries. Often the technique that is employed is to window the time domain response to exclude the reflections and to move the loudspeaker as far away from boundaries as possible during the measurements. The loudspeaker design is a floorstander and the tweeter and midrange will be about 1 m above the floor level. Attempting to directly measure 1m above the ground, at a typical listening distance of 2-3 m and to a sufficiently high resolution and low frequency, would result in contamination of the measurement by ground reflection in the midrange band. For the midrange and tweeter, we can instead elevate these drivers by another meter or more, and measure at a close distance. This will largely eliminate the ground reflection problems for these drivers. On the other hand, our design relies on the ground reflection to augment the woofer panel output. As a result, the measurement on the woofer panel will be carried out on the listening axis at a distance and will include the floor reflection. These two different measurement conditions will subsequently be adjusted to a consistent basis before designing the crossover. I will now explain a somewhat DIY friendly measurement procedure that is based on this principle.

Find an open area outdoors with a flat and relatively smooth surface. I use a long patio area adjacent to my home but the end of a long driveway or the walkway in front of the home can work as well. There must be free space in the “on axis” direction both in front of and behind the loudspeaker of at least 30 feet or more. To the “side” there can be objects or structures as long as they are at least 15 feet away. The dipole nature of the loudspeaker makes it possible to measure closer to structures when they are only to the side of the loudspeaker, since the relative SPL is less there compared to the front, so the energy in the reflection will be weak. The goal is to get a frequency resolution that is around 50 Hz for the midrange and 20-50 Hz for the woofer panel, in order to have sufficient resolution within these bands.

With the loudspeaker positioned as described above, following measurements are conducted:

- Midrange and tweeter: quasi-anechoic measurements are performed on the midrange and tweeter one at a time. The microphone is placed at a known and relatively close distance (e.g. 0.3 to 0.5 m) for these drivers, with the drivers elevated above the ground by 2m or more to minimize the ground plane reflection. Make sure to record the exact distance between the microphone and the plane of the driver’s acoustic center – this information will be needed later. Locate the microphone directly in front of the center of each driver’s diaphragm when making the measurement. Since the acoustic centers have been aligned previously, the microphone stand only needs to be adjusted straight up or down in between measurements.

- Woofer panel: Place the woofer panel on the ground. The microphone is placed on-axis in the far field, at 3m to 4m away from the loudspeaker and elevated to the listening height (e.g. at the intended tweeter height for the final loudspeaker). Record the distance along the ground between the plane of the woofer panel and the plane of the microphone. Choose measurement conditions that will result in a frequency resolution of 50Hz or better and minimal wind and environmental noise contamination. This will ensure good quality low frequency information.

Note that, for each angle on and off axis, a new measurement set must be collected.

Next, additional processing is done to the tweeter and midrange response measurements. The goal is to mimic the propagation of their acoustic outputs from where they were measured, out to where the woofer panel measurements were made. The SPL and phase values are adjusted to account for SPL loss and phase rotation that takes place as sound travels through the air. The SPL adjustment is computed using the equation:

$$\Delta SPL = -20 \log(R_2/R_1)$$

where R_2 is the distance at which the woofer panel measurement was made, and R_1 is the distance at which the midrange and tweeter measurements were made. For example, if the midrange and tweeter are measured at 0.5m and the woofer panel at 3m, ΔSPL is -15.56 dB. This value is added to the SPL level of both the midrange and tweeter to reflect the lower SPL that would have been recorded if those measurement were made at a distance of (in this example) 3m instead of 0.5m. Next, we account for the phase rotation that would have occurred during the time sound was traveling the “extra” 2.5m between the measurements made at 0.5m and 3m. The phase rotation $\Delta\theta$ is equal to $-2\pi df/c$ radians or $-360df/c$ degrees, where c is the speed of sound (343m/sec at 20C at sea level), d the distance in meters, and f the frequency of interest in Hertz. The amount of additional phase rotation is frequency dependent. Because a separate calculation is required for each frequency in the measurement file this is best done using software such as a spreadsheet or using the functionality of the crossover design program itself, and then saving the resulting data to a new file.

Once these adjustments have been made to the measurement data, all measurements have been brought to the same location in space and the crossover design process can begin using crossover modeling software. I will now describe my approach to this task using a DSP (IIR filter) based framework. The nude dipole loudspeaker requires more numerous and different response corrections compared to a boxed loudspeaker. In order to provide the necessary flexibility, the crossover modeling software and the crossover hardware should be able to implement high and low pass filters of first order, of second order with variable Q, parametric EQ, first and second order shelving filters, first and second order all-pass filters, and the biquadratic filter (often referred to commonly as the Linkwitz Transform).

Crossover Development

Crossover development is carried out in two steps: flattening the response for each driver and applying the crossover filters. I typically choose to work on the midrange response first. This driver requires more correction than other drivers because its passband spans the dipole peak. The initial goal is to flatten the frequency response in a region extending well above and below the intended crossover points for this band. This has the additional benefit of flattening the

driver's phase response within the intended passband. At this stage the amount of power lift is not of concern since this will be largely eliminated when the crossover filters are applied if they are of sufficiently high order (e.g. 4th order). The first response area that should be worked on is the region around the dipole peak of the midrange. A combination of a peaking EQ band and a shelf filter work well, although it is a matter of some trial and error on the part of the designer to find the best solution. Additional shaping may be needed to flatten the response sufficiently overall. Note that no crossover filters have been applied so far, only filters that flatten the response. For the woofer panel, the entire passband lies below the dipole peak and flattening its response is a bit more straightforward. First or low-Q second order shelving filters are effective in this regard. Like with the midrange, the goal is to flatten and extend the response of the woofer panel, especially below the intended crossover frequency to the subwoofer. The tweeter typically requires the fewest corrections as the response is often relatively flat in comparison. After the driver responses have been corrected for dipole cancellation and flattened their levels should be made similar via gain adjustment.

At this point the design can progress much like with a boxed loudspeaker, with the goal of achieving a flat system response. Crossover filters should be applied at the target crossover frequencies, e.g. around 350 Hz and 2k Hz, as well as at the desired subwoofer to woofer crossover point. For the low frequency end of the woofer panel, Linkwitz [13] described a useful approach in which the second order highpass near-resonance driver response is transformed via the "Linkwitz transform" biquad equalizer into one stage of the high-pass crossover. For example, if the designer wishes to use a LR4 type crossover at 80 Hz then the driver's F_s and Q can be transformed to $F=80$ Hz, $Q=0.707$ to form one of the second order filter stages that make up the LR4, with the other second order stage implemented with an electronic filter also having $F=80$ Hz, $Q=0.707$, the combination creating the LR4 response. Excess relative phase angle between a pair of drivers at their crossover point may need to be addressed so that the responses will sum correctly. The designer must decide how to continue to flatten and shape the overall system response. The over-arching goal at this stage is a completely flat system response with the drivers operating in-phase at the crossover points.

Voicing the Dipole Loudspeaker

The final stage of crossover development is often called "voicing". The voicing EQ correction is applied to correct the loudspeaker's tonal balance. A flat on axis response from a dipole system will often sound too bright. Research reviewed by Toole [14] and performed by Olive [15] suggests that a down-trending power response over most of the audio band is preferred by both trained and untrained listeners and my own listening tests confirm this. Toole and Olive have published their findings as plots of the steady-state room curves for a variety of highly rated loudspeakers. The in-room steady-state response is similar to the power response of the loudspeaker, which may be estimated by inverting the loudspeaker's directivity index. One source of data regarding the directivity index of loudspeakers can be found online from published Klippel analyses [16]. For many well-regarded loudspeakers the same trend is observed: the directivity index begins to rise above 0 dB between 100 Hz and 300 Hz, reaching 10 dB or more by 10k Hz. Since the loudspeaker's power response can be estimated from the inverse of this curve, the power response is undergoing a 10dB decrease over this frequency range. This power response trend influences how the loudspeaker sounds in the listening space

since the off-axis acoustic energy that is radiated by the loudspeaker reflects from one or more room boundaries and returns to the listening location.

In contrast to a boxed loudspeaker, a nude dipole loudspeaker as designed in the way described in this article will have a flat on-axis frequency response and a flat power response because a dipole has nearly constant directivity. This characteristic makes it easy to tailor the power response and tonal balance by implementing a suitable voicing EQ curve to the crossover input, upstream from the crossover filters. I experimented over a long period of time, using more and less amounts of the down-tilting EQ and changing the range of frequencies in which the voicing EQ was applied. I found that a spectral tilt of approximately 4 to 5 dB per decade provided the most accurate and satisfying tonal balance. This is similar to, but steeper than, a “pinkening filter” used to convert white noise to pink noise. The down-tilt is used between 100 Hz and 10k Hz, typically amounting to 8dB to 10dB of correction in total. At the highest frequencies, the rising directivity of the tweeter will continue the trend without requiring EQ. [17, Figure 4.20] One way to implement this type of correction in DSP is via a filter bank consisting of multiple first order shelf filters, each applying only 1-2 dB of correction. Design of similar analog active pinkening filters is discussed by Self in reference 18, section. 11.14 “equalizers with non-6 dB Slopes”. For a passive crossover network, the spectral tilt should be targeted during the initial crossover development process instead of as a separate voicing EQ in addition to a flat response. At this point the crossover development is complete and the system is ready for audition.

Anytime a trial crossover is undergoing evaluation, several listening sessions should ensue using program material that is familiar to the designer. An iterative process of listening, tweaking and adjusting of the voicing EQ correction should be used. At this point the system should be very close to being well balanced tonally, and only very minor adjustments should be necessary. There are many small clues to listen for, but it is important to keep in mind that the tonal balance in the recording was set during the mix, and this was judged using loudspeakers with their own tonal balance. This creates what is known as the circle of confusion, a term coined by Floyd Toole [19, pp.18-24]. Because there is no established standard it is often left to chance that the monitors at the mixing console and the immediate acoustic environment were tonally neutral and did not imprint their own sound upon the recording. Therefore, listening to a range of high-quality music from various sources and styles is essential during the evaluation. An astute and experienced listener can often detect whether the entire recording is tonally off-balance and make judgements accordingly. It helps if the critical listening is spread out over several days and lasts only a couple hours at a time, with several hours or more of downtime to give the ears a rest. Repeat the listening evaluation in the morning and you will be rewarded with a fresh perspective. In the nude wireframe dipole loudspeaker, the voicing process is relatively straightforward because it consists only of optimizing the voicing equalization curve and choosing the subwoofer level.

The development of a minimally-baffled nude dipole loudspeaker, when executed as described above, has a similar feel each time. This may be the result of several factors. Because the size of the midrange falls within a narrow range its passband always spans between approximately 350 Hz and 2k Hz. The response of the midrange is similar from driver to driver, and each is distinguished more by the off-axis performance above 1k Hz and the distortion performance overall. The choice of driver used in the woofer panel is more liberal, and primarily the distortion below 200 Hz and ability to operate cleanly up to and above the midrange crossover point are the

characteristics that differentiate one implementation from another. The crossover and voicing equalization are similar from system to system, and this familiarity streamlines the crafting of the loudspeaker as the designer gains experience.

An Example of the Wireframe Dipole Loudspeaker Concept

An example of the minimally baffled dipole loudspeaker is now described. Drivers were initially screened using measurements performed on- and off-axis to establish the range over which they were suitable for a dipole application (see for example Figure 4 and Figure 7). This is most critical for the midrange, which must be use both above and below the dipole peak, followed by the tweeter. Because the woofers are used primarily at relatively low frequency only the driver's distortion performance, as measured by others [20], and its Thiele-Small parameters were considered as selection criteria.

The drivers used in the loudspeaker include:

- Aurum Cantus AST2560 tweeter with the faceplate removed
- SB Acoustics SB17MFC-4
- Four Peerless 830869 HDS woofers in a 20x20 inch baffle.

These were paired with a set of closed box subwoofers that were already present in the listening space. This prototype loudspeaker is shown in Figure 9.

The acoustical measurement software ARTA [21] was used in dual-channel mode for all measurements. Measurements were conducted outdoors, on axis only, following the procedure described earlier with the drivers elevated above the ground by approximately 2.5 m and a mic distance of 0.5 m. The woofer panel was measured separately at approximately 4 m distance, and at the intended listening height for the completed loudspeaker. Subsequently the data for the midrange and tweeter were brought out to the measurement distance at which the woofer panel was measured, bringing all measurements to the same point in space.

Crossover design was performed using the ACD design tools written by the author [22]. The DSP crossover was implemented using software-based IIR filters on a computer running Linux and connected to a Behringer UMC1820 audio interface. Software included the LADSPA plugin ACDf written by the author [23], with routing and processing tasks handled by a Gstreamer pipeline [24] run from the command line [25]. A list of ACDf filter types is provided in Table 1. This should be consulted when reading subsequent lists of crossover and voicing EQ filters, given in Tables 2-5. All first and second order filters are defined in terms of their gain (dB), polarity, pole frequency (F_p), pole quality factor (Q_p), zero frequency (F_z), and zero quality factor (Q_z). The polarity value 1 indicates normal polarity, with any other value used to indicate reverse polarity. Not every parameter is used for every filter type.

As mentioned earlier in the text, I like to work on the midrange band first. The SB17MFC is one of the easiest midrange drivers to use in this application because of a lack of response anomalies in the dipole peak area and its relatively low distortion. I have found that a correction consisting of an EQ band and a second order low Q shelving filter is a good starting point to flatten out the response. The goal is to make the response flat across as broad a band as possible. This typically means up to about where the cone breakup begins, and at least 1 octave below the intended crossover point. As an example, the midrange raw response is shown in Figure 10 along with the

flattened response resulting from these initial corrections. In this case I used a biquadratic filter (type 28) to implement the shelf, since this permits different Q values for the start and end of the transition, and a PEQ band (type 26) having $Q=2$ at 1.6k Hz to address the dipole peak. One additional other filter was used; a second order shelving filter (type 25) further flattens the response above 2k Hz. The response comprising these “passband flattening” filters is shown in Figure 11. Because 4th order crossover filters will be used, the amount of low frequency power lift will be strongly diminished below the eventual crossover point. With the response flattened, LR4 type crossover filters were added in the usual way. The resulting response is shown in Figure 12. The complete set of filters used for the midrange band are listed in Table 3.

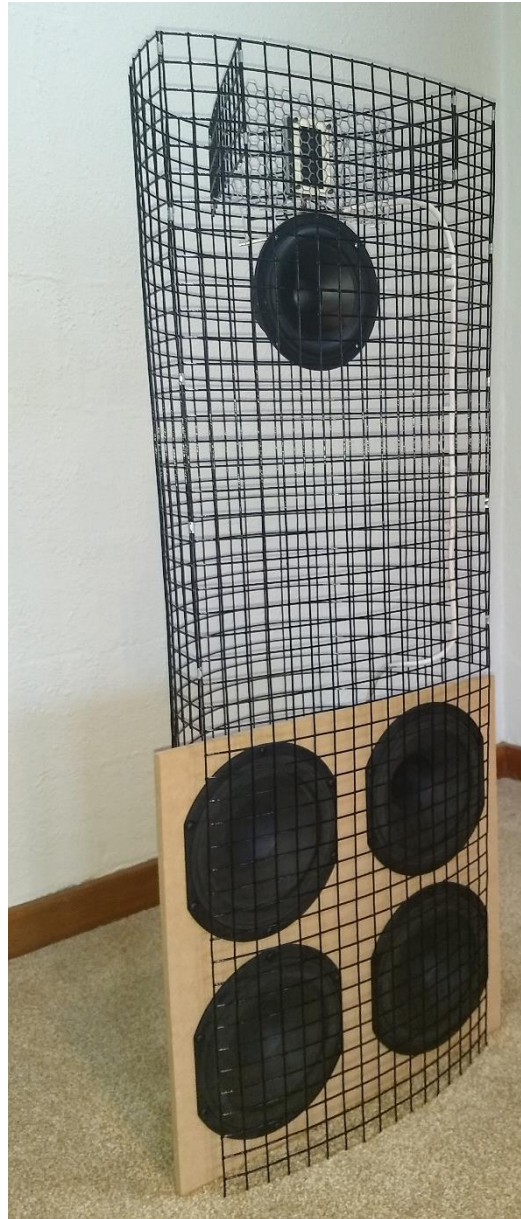


Figure 9 : The prototype loudspeaker.

The response from the minimally baffled tweeter is the easiest to work with in this loudspeaker. A shelving filter was used to correct a response droop below 3k Hz. Thereafter, high pass filtering complementary to that used in the midrange was created. This was implemented by the combination of a biquadratic filter (type 28) and a second order high-pass filter (type 22). An all-pass delay was used to improve the phase angle match to the midrange at the crossover point. The raw, as measured tweeter response and the final response with crossover filters applied is shown in Figure 13. The filters used for this band are listed in Table 2.

Figure 14 shows the measured response of the woofer panel and the corrected response with crossover filters applied. The Peerless 830869 provides low distortion down to 80 Hz and has an open basket. With four closely spaced drivers on a minimal baffle, the panel operates much like one larger driver at low and middle frequencies. Above 500 Hz or so, there is an increasing amount of driver-driver interaction that causes nulls to form. In general, the panel has a dipole-like response, with a dipole peak around 300 Hz. A first-order shelf (filter type 4) was used to compensate for the falling response below 300 Hz. This was all that was needed to flatten out the response below the dipole peak. A second order low pass filter was used to define the upper end of the passband, targeting a 350 Hz crossover point. This combines with the existing roll-off above the dipole peak to produce a higher effective slope that can match up to the 4th order crossover filters used to define the low end of the midrange passband. To define the low end of the woofer passband, two filters were used. A biquadratic filter was used to transform the near-resonance response to $Q=0.707$ and $F=80$ Hz. This forms one part of the LR4 crossover filter. The other part is a second order high-pass filter, also with $Q=0.707$ at 80 Hz. Because of the large amount of gain used in the first order shelving filter, there is an overall gain correction (filter type 0) of -16.25 dB. The filters used for this band are listed in Table 4.

The system response (not including the subwoofer) before implementing the voicing EQ is shown in Figure 15, with the crossover filter responses for tweeter, midrange, and woofer panel shown in Figure 16. By choosing a flat response target, the task of getting the drivers well integrated and the levels balanced is relatively straightforward. This also decouples the task of integrating the drivers into a system from the task of voicing the loudspeaker. Because the measurement data is not extremely clean, there is some ambiguity to the exact levels of each driver, but in general the system response is approximately ± 1 dB between 150 Hz and 10k Hz.

The voicing EQ curve that was employed for this design is shown in Figure 17 and the shelving filters listed in Table 5. This consists of a down-tilted trend of approximately 5dB per decade, resulting in a 9dB difference in the level at low and high frequencies. The voicing EQ was generated via the method discussed earlier and was tuned by ear using both familiar and unfamiliar program material over a period of about 2 weeks.

As mentioned earlier, the dipole loudspeaker was integrated with an existing subwoofer system. This consisted primarily of setting the appropriate level for the subwoofer. The subwoofer system consists of multiple separate 12 inch woofers, each in a sealed, 45 liter enclosure, placed against the front wall behind the dipole loudspeakers. These use a “Linkwitz Transform” biquadratic filter to alter their low frequency behavior and match their response to the room gain. The transformed response of the subwoofers was set to $F=25$ Hz, $Q=0.6$ and an 80 Hz LR4 crossover was used to match them to the main speakers.

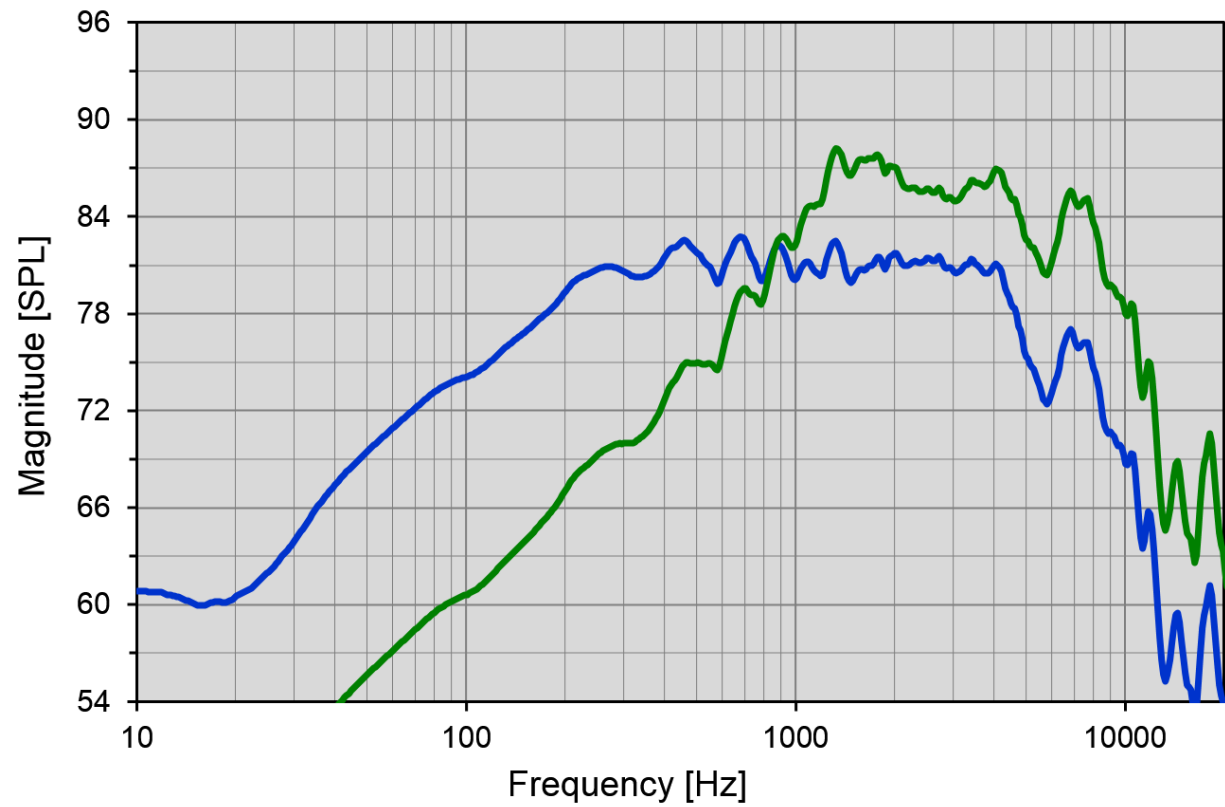


Figure 10: Midrange responses; raw (green) and after applying filters to flatten the response (blue).

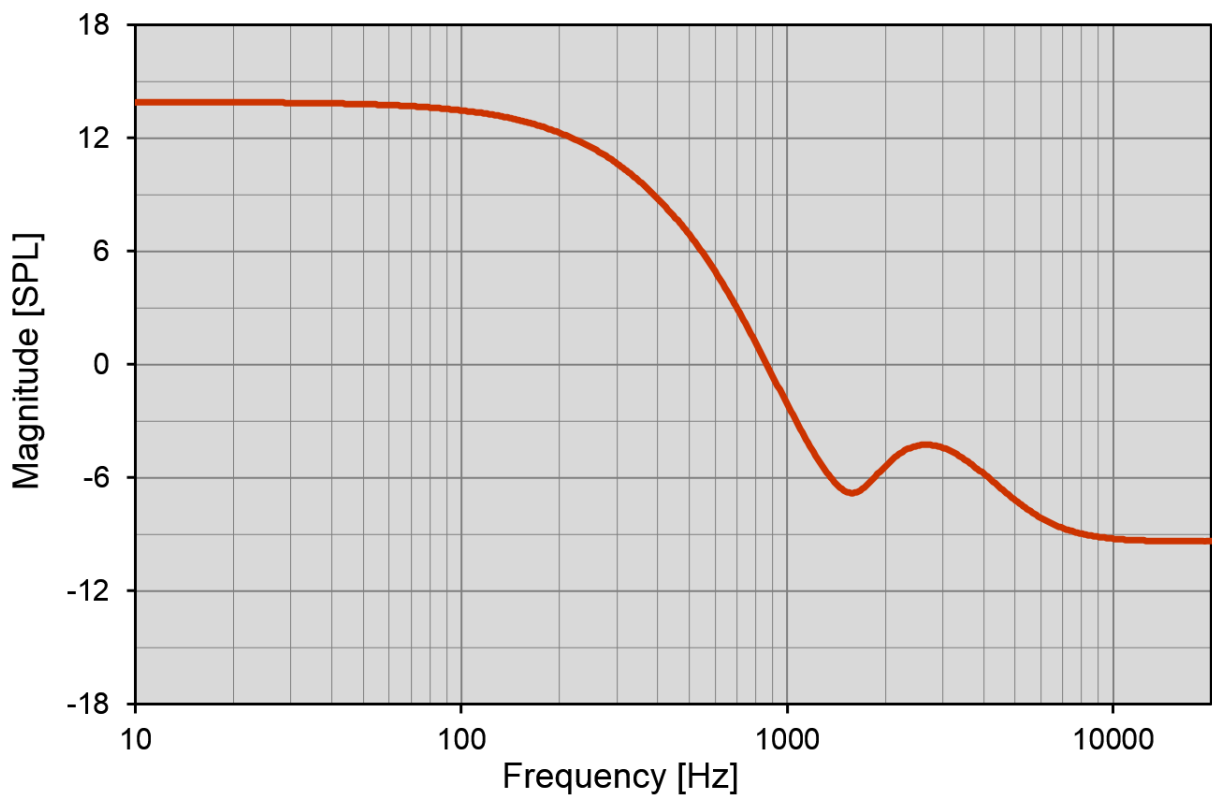


Figure 11: Response of the filters used to flatten the raw midrange response.

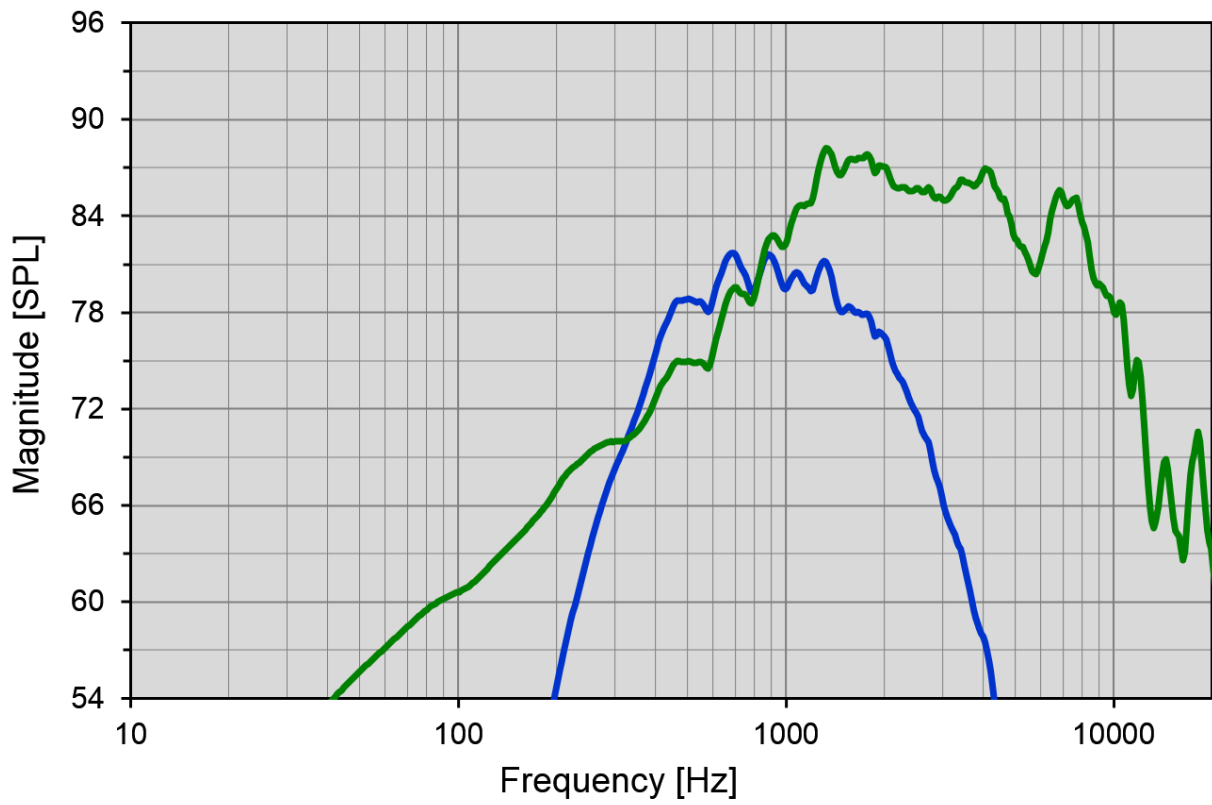


Figure 12: Midrange responses; raw (green) and with crossover filters applied (blue).

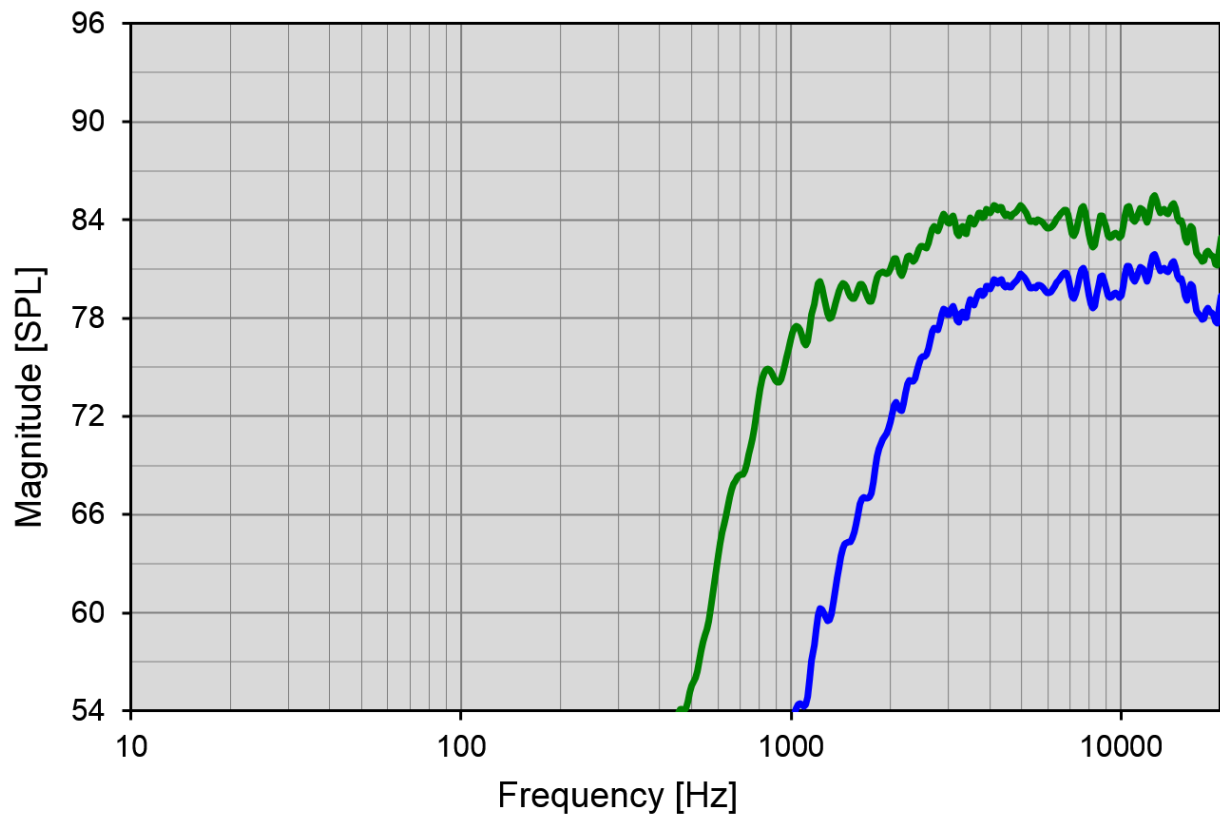


Figure 13: Tweeter responses; raw (green) and with crossover filters applied (blue).

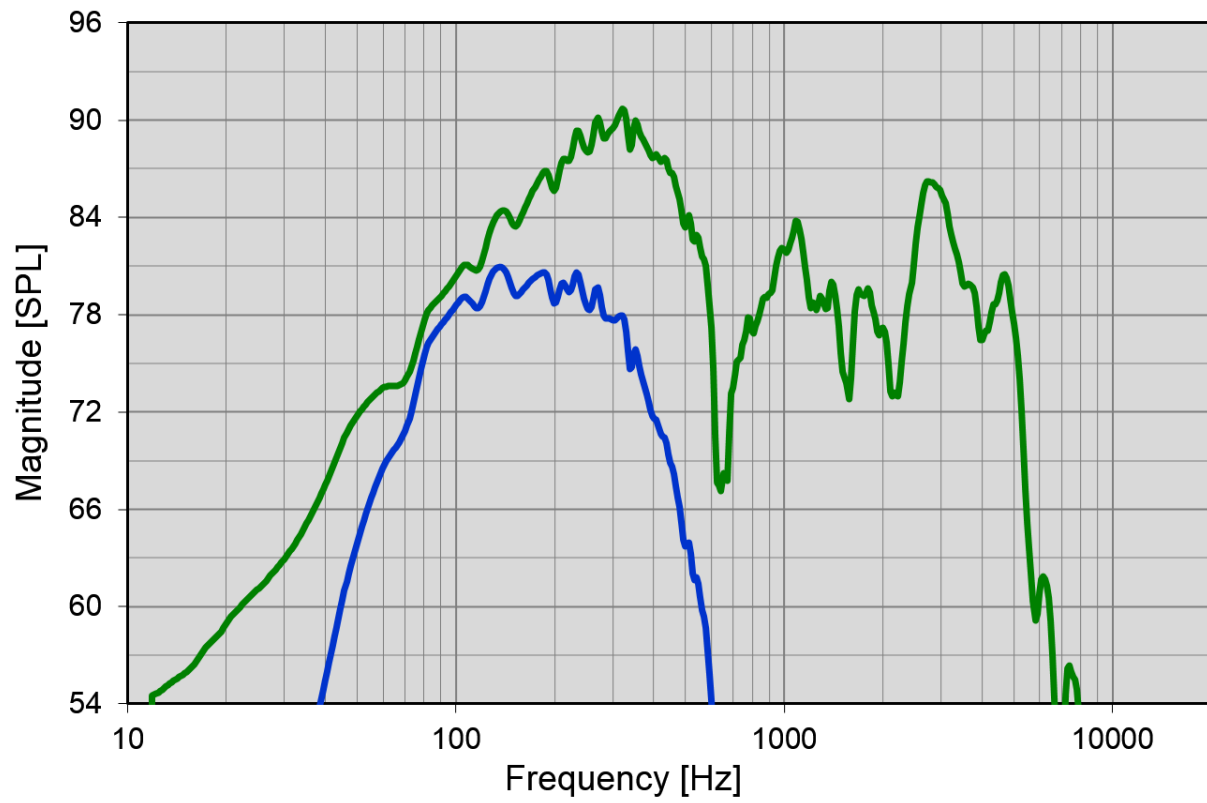


Figure 14: Woofer responses; raw (green) and with crossover filters applied (blue).

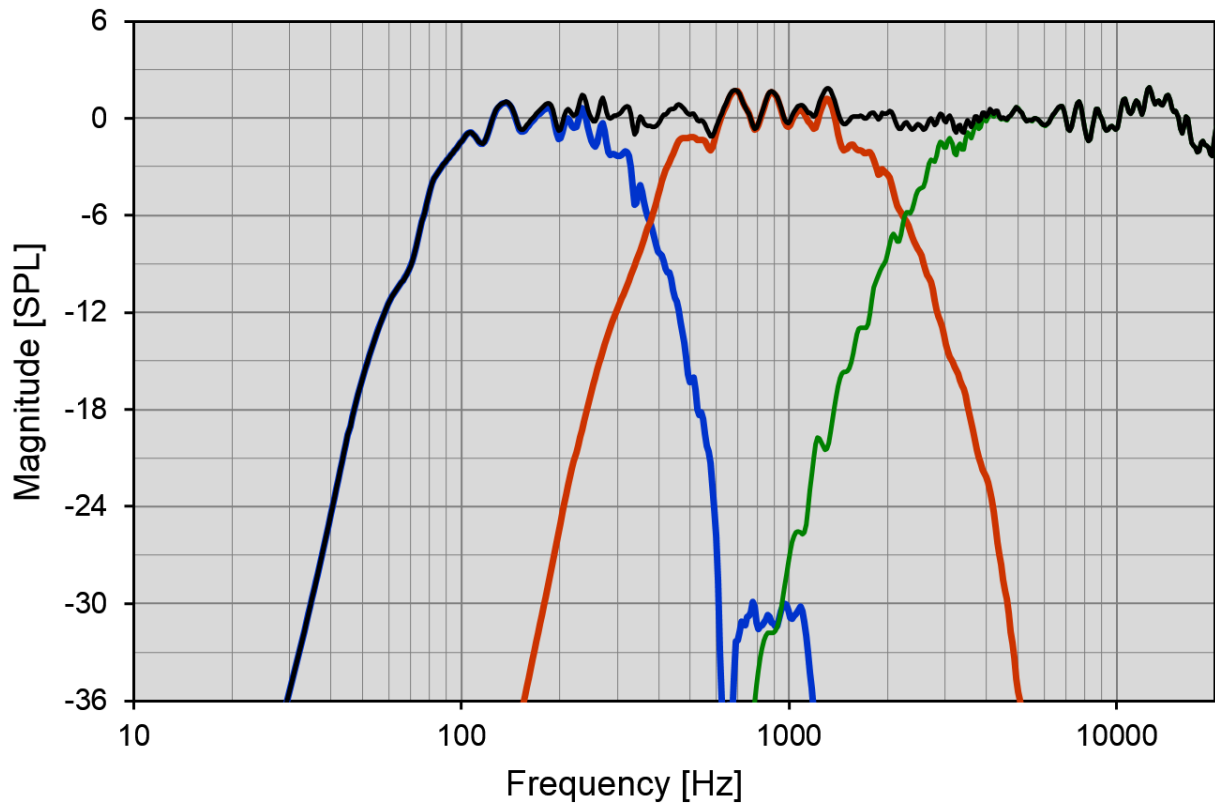


Figure 15: Responses after the crossover filters have been applied; woofer (blue), midrange (red), tweeter (green), and the combined system response (black).

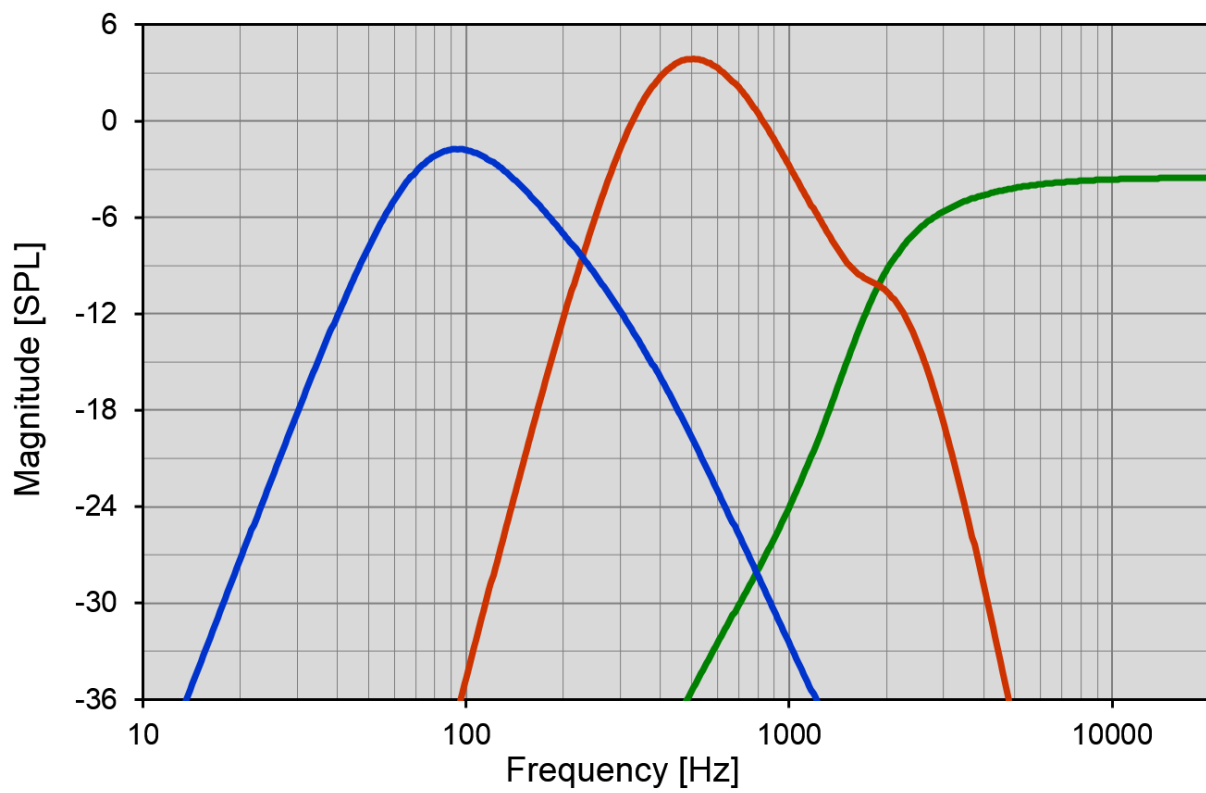


Figure 16: Responses of crossover filters for woofer (blue), midrange (red), and tweeter (green).

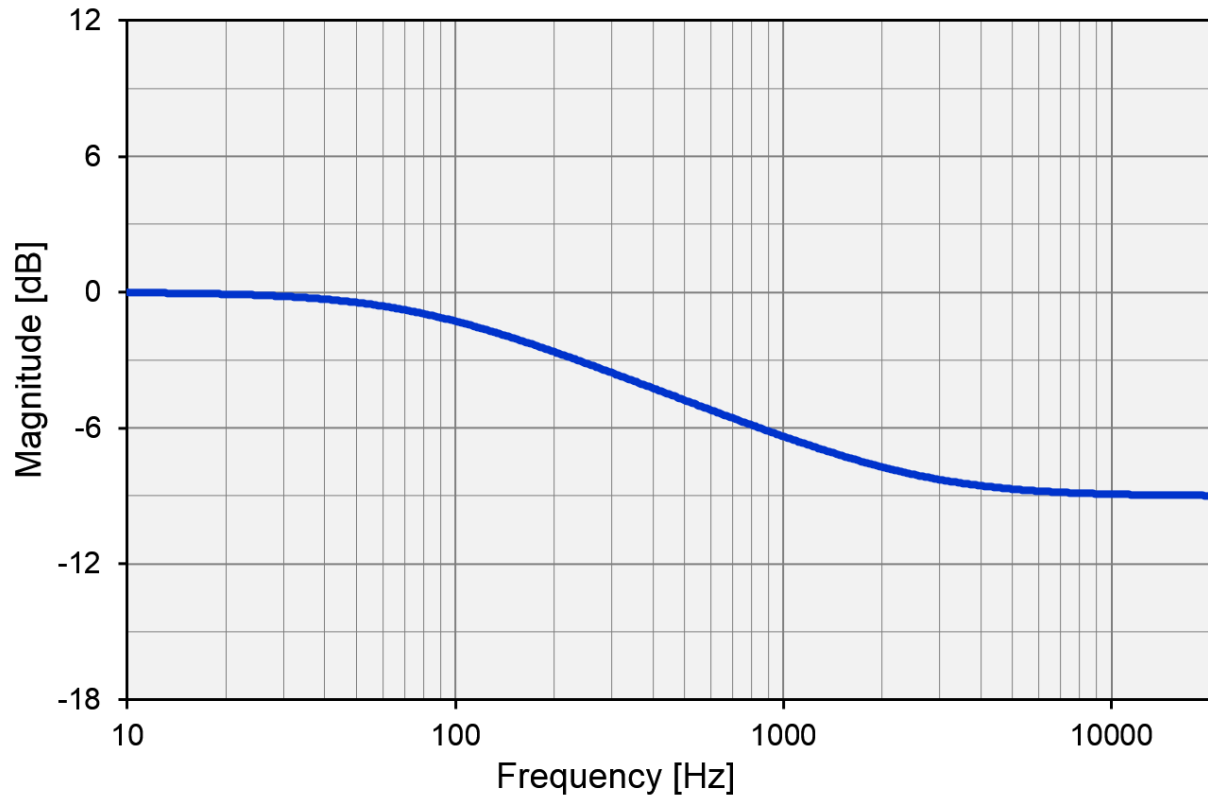


Figure 17: Response of the voicing EQ filter.

Table 1: A description and the required parameters for each filter type.

TYPE	DESCRIPTION	REQUIRED PARAMETERS
0	gain block	db, polarity
1	1st order LP	db, polarity, Fp
2	1st order HP	db, polarity, Fp
3	1st order AP	polarity, Fp
4	1st low shelf	Fp, dB
5	1st high shelf	Fp, dB
21	2nd order LP	dB, polarity, Fp, Qp,
22	2nd order HP	dB, polarity, Fp, Qp
23	2nd order AP	polarity, Fp, Qp
24	2nd order low shelf	dB, polarity, Fp, Qp
25	2nd order high shelf	dB, polarity, Fp, Qp
26	parametric EQ	dB, Fp, Qp
27	2nd order notch	dB, polarity, Fp, Qp, Fz
28	biquadratic filter	dB, polarity, Fp, Qp, Fz, Qz

Table 2: A list of filters applied to the tweeter band

Filter/EQ:	1	2	3	4	5
type:	0	24	28	22	23
gain [dB]:	-3.5	5	0	0	0
polarity:	1	1	1	1	1
Fp:		2200	2200	2200	6500
Qp:		0.85	0.707	0.707	0.56
Fz:			1200		
Qz:			1		

Table 3: A list of filters applied to the midrange band

Filter/EQ:	1	2	3	4	5	6	7	8
type:	0	28	26	25	22	22	21	21
gain [dB]:	-0.35	0	-3	-9	0	0	0	0
polarity:		1	1	1	1	1	1	1
Fp:		550	1600	4000	400	400	2100	2100
Qp:		0.45	2	0.7	0.707	0.707	0.707	0.707
Fz:		1250						
Qz:		1.1						

Table 4: A list of filters applied to the woofer band

Filter/EQ:	1	2	3	4	5
type:	0	4	28	22	21
gain [dB]:	-16.25	24	0	0	0
polarity:		1	1	1	1
Fp:		150	80	80	375
Qp:			0.707	0.707	0.6
Fz:			30		
Qz:			0.3		

Table 5: A list of filters employed as Voicing EQ

Filter/EQ:	1	2	3	4	5	6
type:	5	5	5	5	5	5
gain [dB]:	-1.5	-1.5	-1.5	-1.5	-1.5	-1.5
polarity:	1	1	1	1	1	1
Fp:	100	182	331	603	1098	2000
Qp:						
Fz:						
Qz:						

One question that is often asked is “what is the sensitivity of the loudspeaker?”. It is not straightforward to answer this question, because most drivers in the loudspeaker do not have a

flat passband and the power delivered to each driver changes significantly with frequency. At the same time, it is useful to make some estimation of the sensitivity to understand how much amplifier power will be appropriate. To do this we can start with the initial “flat on axis” design target before the voicing EQ was applied, since all drivers have been made to operate at the same SPL via the crossover. The tweeter has a rated sensitivity (by the manufacturer) of 95 dB/W/1m and the dipole response is relatively flat above 2k Hz. If we look at the crossover filters shown in Figure 16 we can see that the tweeter is attenuated by 3.5 dB to match the level of other drivers. This suggests that the system sensitivity is approximately 91 dB/W/1m. Looking again at Figure 16, we see that the midrange filter peaks at +3.75 dB at approximately 525 Hz. This reduces the sensitivity between approximately 300 and 850 Hz to 87 dB/W, with the sensitivity returning to the 91 dB/W level for other frequencies. This relatively high sensitivity results from operating the midrange over a limited band near the dipole peak, using a combination of electrical and ground plane acoustic gains for the woofer panel, and restricting the low frequency for the woofer panel to 80 Hz to limit the amount of low frequency power lift that is required.

In order to assess the relative power requirement for each band we must also consider the effect of the down-tilting voicing EQ. To capture both the power demand across each band as prescribed by the crossover filters as well as the voicing EQ these were added together on a logarithmic basis. Figure 18 shows the resulting frequency dependence of the input power requirement. The voicing EQ has reduced the midrange band power demand in the vicinity of 525 Hz to below 0dB. We can also see in Figure 18 that the input power demand on the tweeter is 12 dB (16 times) lower than for the other bands, which is a significant difference. Because the loudspeaker system is active, the amplification power can be appropriately chosen for each band and identical amplifiers are not required as long as any difference in gain between amplifiers is accounted for.

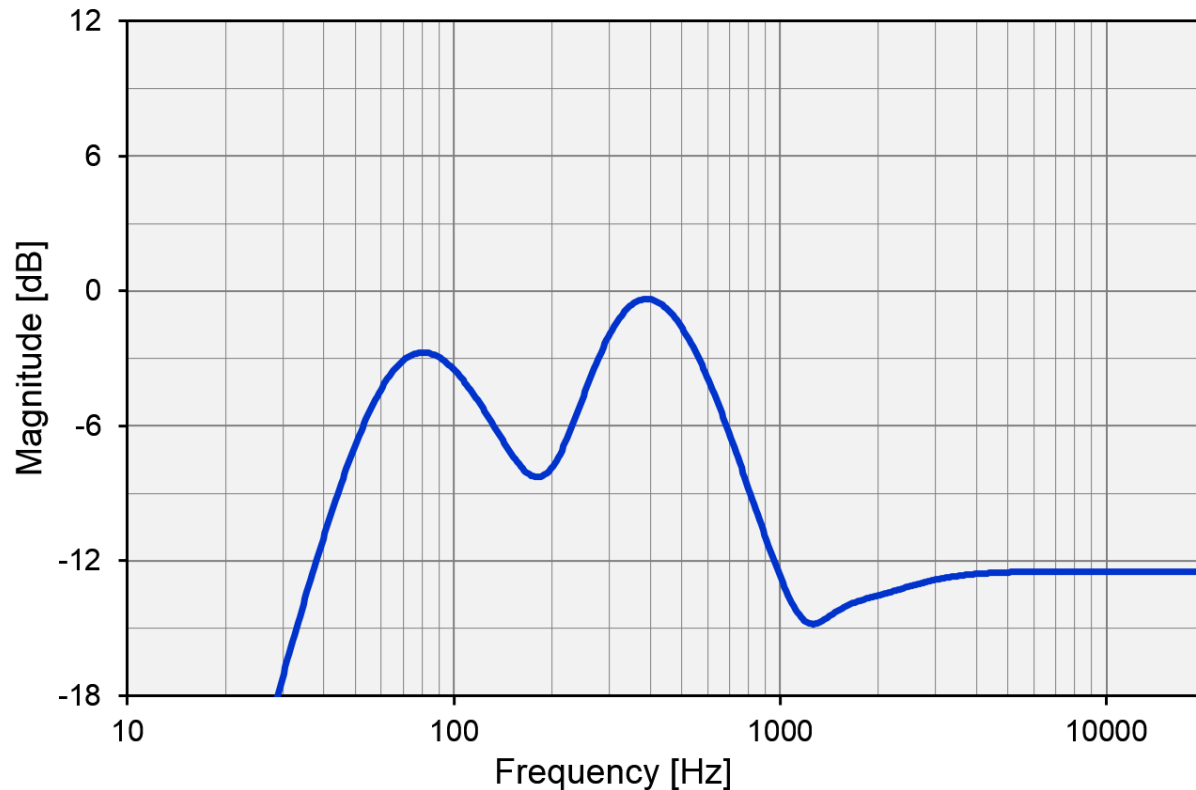


Figure 18: Relative input power requirement for the loudspeaker system as a function of frequency.

Listening Impressions

The loudspeaker was set up in the main listening area - a large, wide space with a vaulted ceiling. The front wall is a fireplace with a wide river rock face and a stone hearth. Behind the listener the space continues into the kitchen above a counter with cabinets below. Left and right subwoofers flank the fireplace in a 4-high stack on either side and are driven by separate amplifiers. A sofa is located at the listening location. The loudspeakers were separated by about 2.2 m and placed approximately 1m in front of the fireplace face, which protrudes out from the front wall by approximately 0.5 m. The listening location is approximately 2.5 m from the speakers with the ear on axis with the tweeter, the center of which is 1.1 m above the floor. The system was powered via amplification of approximately 100W into 8 Ohms per channel, with identical amplifiers used on the woofer, midrange, and tweeter bands.

Well, how do they sound? I feel that they invoke a sense of a really large loudspeaker but without any of the associated coloration - it's a monkey-coffin speaker without the box resonance issues and with better imaging. The midbass especially sounds to me very clean and unrestrained. There is a large soundstage. On a live recording with sufficient location cues you really get a sense of the space where the performance took place, with sounds coming from beyond the speakers, both to the side and above. The dynamics seem very unconstrained. What is very striking is that the tonal balance remains almost exactly the same no matter where you go in the room. There is a chair in the corner of the room, about 10-12 feet to the side past one speaker

and slightly behind it. I often sit here to drink my coffee in the morning. If I get up from the main, central listening spot, walk to the corner chair and sit down, at no time does the tonal balance change. The stereo image is not formed at this chair, being so far off to one side, but the tonal balance sounds the same, save the bass which is slightly elevated there (due to its close proximity to the subwoofers). Even more astounding is walking from the central listening spot forward until you pass directly between the loudspeakers. The stereo image along that route only gets better. Directly between the loudspeakers the listener is located in the 90-degree dipole “null”. Because the sources are not point-like, it’s not a perfect null but might approach 20 dB deep. Primarily the sound being heard is coming from the front wall reflection. Because the material is stone, it can reflect all frequencies more or less equally, and the rear output has identical frequency response as the front. The sound stage is like huge headphones: the image is diffuse but immersive and expansive. It’s a unique experience. Back at the central listening location, the stereo image is similar to what one gets from a high-quality mini-monitor, which the baffle-less midrange and tweeter approximate. Overall, the loudspeaker provides a very satisfying musical experience, not just at the central listening location but in different parts of the home wherever I can perceive their output. This is likely due to the fact that the power response has been shaped to be like the typical boxed loudspeaker while the dipole radiation character projects the same tonal footprint in all directions.

I hope that the reader now has some interest in building a minimally baffled dipole loudspeaker themselves. What I have described in the prototype loudspeaker example above is one possible embodiment. As long as the “minimal baffle” paradigm is followed there may be other ways to create the same type of loudspeaker. Some improvements could be undertaken as well, such as decoupling the woofers from the planar “woofer panel”, since it is not the sound pressure in the air but the vibration the motor induces via the frame into the panel that leads to panel resonances. Another method, apart from the wireframe scaffold used in this embodiment, could provide support, decoupling, and a more aesthetically pleasing enclosure for the midrange and tweeter. I encourage builders to experiment and to make careful measurements when constructing their own version of this wonderful loudspeaker.

Acknowledgements: I would like to extend my sincere thanks to Tim Mellow for sharing data arising from his investigation into baffled dipole and monopole sources. This made it possible to construct several useful figures in this article. Also, I would like to thank Andrew Jones for spurring the analysis of the sound pressure losses for a freely hanging driver.

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